

SCHOLARSHIP RECORD MANAGEMENT WITH ELIGIBILITY  
RECOMMENDER FOR HIGHER EDUCATION INSTITUTION

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## **BACHELOR OF SCIENCE IN COMPUTER SCIENCE**

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## **ABSTRACT**

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## **CHAPTER I**

### **INTRODUCTION**

#### **Program Context**

Scholarship programs play a vital role in enabling students to pursue higher education. They significantly reduce financial burdens and support students' right to quality education (Morato, 2022). In the Philippines, scholarship and financial aid programs within State Universities and Colleges (SUCs) serve as essential mechanisms to promote equitable access to higher learning (Remegio, 2025). According to Capinig et al. (2023), scholarship grantees express satisfaction with the programs under the Commission on Higher Education (CHED), noting that subsidies ease their educational expenses and reduce stress.

At Biliran Province State University (BiPSU), scholarship programs are crucial in providing financial assistance to academically deserving and economically challenged students. The university offers internal scholarships such as Academic Scholarships for College and University Scholars, as well as institutional programs for BiPSU employees called BiPSU Staff Scholarship. BiPSU also partners with external agencies to implement scholarship initiatives such as the DOST Science and Technology (S&T) Undergraduate Scholarships, the Junior Level Science Scholarship (JLSS), and CHED programs including the CHED Merit Scholarship Program (CMSP) and Coconut Scholarship (CoScho). National government programs like the Tertiary Education Subsidy (TES) and the Tulong Dunong Program (TDP) contribute millions of pesos in financial support annually, as reported by the

Philippine Information Agency (2022). Additionally, BiPSU implements special initiatives such as affirmative action scholarships for Indigenous Peoples (IP's), Persons with Disabilities (PWD), and marginalized groups like students from depressed areas, demonstrating its commitment to inclusive higher education (Arinto, 2024).

Despite these opportunities, the absence of a centralized scholarship database at BiPSU results in uneven information distribution, delays in verification, and inconsistencies in the evaluation process. Interviews with the Staff Office of the Student Development and Services Office (SDSO) and the UniFAST Office reveal that scholarship applications are still processed manually through physical submissions. Documents are stored in folders and filing cabinets, making processes slow, tedious, and susceptible to human error. Bagaporo et al. (2025) explain that manual, paper-based systems are time-consuming, vulnerable to data loss, and prone to inaccuracies such as typographical errors and missing records. Agu et al., (2022) similarly emphasize that manual recordkeeping leads to misplaced files, poor retrieval efficiency, and administrative delays that negatively affect service delivery.

These challenges become more evident when staff prepare reports for partner agencies such as the Department of Science and Technology (DOST) and the Commission on Higher Education (CHED). Staff members must manually search through stacks of documents, resulting in prolonged processing times, disorganized data, and heavy workloads. Retrieving older records is especially difficult because physical storage systems do not support efficient searching. Duranti (2001) notes that paper-based archives are inefficient and unreliable, whereas digital archiving enhances accuracy,



accessibility, and preservation. Emerging research also shows how artificial intelligence and rule-based filtering can improve archival workflows, automate classification, and accelerate data retrieval (Chao et al., 2023). Without digital tools, BiPSU is unable to maximize intelligent processes for verifying applicant,, identifying duplicate scholars, or determining scholarship eligibility through structured data comparisons.

The current manual system also causes operational problems such as unequal dissemination of scholarship announcements, inconsistent records, and potential duplication of scholarship benefits. The absence of standardized criteria and automated verification raises concerns regarding fairness, transparency, and the possibility of favoritism or a “backer system.” These inefficiencies limit the ability of the Student Development and Services Office (SDSO) to provide timely guidance and personalized mentorship to students seeking scholarships.

Given these challenges, the need for a modernized, automated, and data-driven scholarship management system becomes evident. Prior research highlights the effectiveness of digital systems in improving efficiency, accuracy, and transparency within administrative processes (Duranti, 2001; Acebro, 2024). Ricci et al. (2015) further emphasize the value of recommender systems that use Rule-Based Filtering to match user profiles with relevant opportunities. Building on these concepts, this study proposes the development of a Scholarship Record Management with Eligibility Recommender for Higher Educational Institution like BiPSU. The system will consolidate all scholarship programs, requirements, student and employee data into a single platform that supports fast record retrieval, automated

eligibility checking, transparent evaluation, and equitable scholarship distribution. By using Rule-Based Filtering, the system can match student and employee attributes to scholarship requirements and eligibility criteria, and generate personalized recommendations.

Integrated within this modernization effort is the study's support of several United Nations Sustainable Development Goals (SDGs). The system directly aligns with SDG 4 (Quality Education) by promoting equitable access to educational opportunities through transparent and consistent scholarship evaluation. By offering an accessible platform where students can view scholarship programs and receive personalized recommendations, the system reduces information gaps and strengthens opportunities for financial aid, supporting claims that digital scholarship systems increase transparency and equal access (Demirkan & Delen, 2012). It also advances SDG 10 (Reduced Inequalities) by minimizing bias, preventing duplicate benefits, and reducing the influence of personal connections or "backer systems." Automated, data-driven recommender systems reduce human subjectivity and promote fairness in scholarship distribution (Jin et al., 2023b). Furthermore, the development of a modern, web-based scholarship management platform supports SDG 9 (Industry, Innovation, and Infrastructure) by replacing inefficient manual workflows with innovative digital tools that enhance data management and institutional efficiency. The integration of intelligent data processing and recommender algorithms strengthens digital infrastructure in higher education institutions (Ricci et al., 2015c). Finally, the system contributes to SDG 16 (Peace, Justice, and Strong Institutions) by promoting transparency, accountability, and fairness in scholarship processes.

Standardized evaluation criteria, automated filtering, and audit trails help prevent favoritism or manipulation, enhancing institutional trust and integrity (Cordella & Paletti, 2019).

Through the automation of scholarship processes and the integration of data-driven recommendation features, the proposed system aims to eliminate duplicate grants, reduce administrative burdens, improve evaluation consistency, and enhance transparency. Most importantly, it ensures that all students have equal access to scholarship programs and receive guidance based solely on their qualifications. The development of this centralized and intelligent scholarship management system aligns with broader digital transformation initiatives in higher education and reinforces BiPSU's commitment to fair, inclusive, and efficient student support services.

### **Purpose and Description**

The purpose of this study is to develop a Scholarship Record Management with Eligibility Recommender for Higher Education Institution like Biliran Province State University – Main Campus. The system aims to automate the processing, storage, retrieval, and evaluation of scholarship information while providing students and administrators with personalized scholarship recommendations using Rule-Based Filtering. This innovation addresses the current issues caused by manual, paper-based processes in the Office of the Student Development and Services Office and UniFAST Office, such as slow retrieval of records, lack of system that analyzed scholarship data, duplicated scholarship benefits, inconsistent evaluation practices, and unequal access to scholarship announcements.

The system will serve as a centralized digital platform that consolidates all internal and external scholarship programs, their respective requirements, and student and employee profiles. It enables automated eligibility checking, transparent evaluation, faster data retrieval, and fairer scholarship distribution. Through its recommender feature, the system supports both students, employees, and administrators in discovering suitable student for a specific scholarships based on their qualifications, academic performance, and personal information. It also strengthens the SDSO Staff and UniFAST Office's capacity to identify deserving students by providing accurate and data-driven scholarship eligibility recommendations.

Upon the completion of this project, the following individuals and groups are expected to benefit:

*Student Development and Services Offices & UniFAST Office.* The office will benefit from a centralized, automated system that reduces manual work, prevents duplication of scholars, ensures consistent evaluation, and allows fast retrieval of records and reports. The recommender feature assists SDSO and UniFAST personnel in guiding and mentoring students more effectively.

*BiPSU Scholar and Non-Scholar Students.* Students of Biliran Province State University (BiPSU) will have equal access to available scholarship programs, along with personalized recommendations based on predefined scholarship eligibility criteria. The system promotes transparency, minimizes favoritism, and increases students' opportunities to identify scholarships for which they are qualified.

*BiPSU Regular Employees.* Regular employees who are eligible for the BiPSU Staff Scholarship will benefit from faster verification of their eligibility, streamlined application processes, and improved access to scholarship information. The system ensures that qualified staff are accurately identified and supported through automated evaluation and organized record-keeping.

*Scholarship Providers (Internal and External).* Partner agencies and scholarship sponsors will benefit from more accurate and organized data reporting. The system ensures that only qualified and non-duplicated applicants are endorsed.

*University Administrators.* University administrators can utilize the system as a decision-support tool to monitor scholarship distribution, evaluate program utilization, and formulate policies aimed at improving student financial assistance. The report generation module will also support institutional reporting requirements, accreditation processes such as the Certificate of Program Compliance (COPC) and the Accrediting Agency of Chartered Colleges and Universities in the Philippines (AACCCUP), as well as international reporting requirements for the Times Higher Education (THE) Rankings and the World University Rankings for Innovation (WURI).

*Future Researchers and Developers.* The system will provide a practical reference for integrating content-based filtering, cosine similarity, and web-based information systems in scholarship management. It can serve as a foundation for future innovations in academic decision-support systems.

## **Objectives of the Study**

The main objective of this study is to develop a Scholarship Record Management with Eligibility Recommender for Higher Education Institution.

Specifically, it seeks to attain the following:

1. Develop a web-based application that enables BiPSU students, employee, and administrators, in terms of:
  - 1.1. viewing scholarship program informations;
  - 1.2. managing student scholars profiling and record archiving;
  - 1.3. viewing statistical summaries and data of previous and current scholarship records;
  - 1.4. ranking students and BiPSU educators based on scholarship eligibility criteria.
2. Integrate a rule-based filtering scholarship eligibility recommender that provides ranked list of eligible students and employees for the following scholarship:
  - 2.1. Affirmative Scholarship;
  - 2.2. BiPSU Staff Scholarship;
  - 2.3. Tertiary Education Subsidy (TES).
3. Develop a report generation module that automates the creation of scholarship-related reports for institutional, accreditation, and international reporting of SDSO and UniFAST Office requirements including:
  - 3.1. Student Development and Services Office:

- 3.1.1. List of scholars for internal and external scholarships;
  - 3.1.2. summary of consolidated scholars for internal and external scholarships;
- 3.2. UniFAST Office:
  - 3.2.1. list of TDP grantees subject for validation;
  - 3.2.2. list of continuing TES grantees subject for validation;
  - 3.2.3. consolidated continuing TES grantees details;
- 4. Evaluate the developed web-based system using white box testing and the ISO 25010 or International Software Quality Standard in terms:
  - 4.1. Functional Suitability;
  - 4.2. Performance Efficiency;
  - 4.3. Compatibility;
  - 4.4. Interaction Capability;
  - 4.5. Reliability;
  - 4.6. Security;
  - 4.7. Maintainability;
  - 4.8. Flexibility;
  - 4.9. Safety

## **Scope and Limitations**

This study focuses on the development of a Scholarship Record Management with Eligibility Recommender System for Biliran Province State University (BiPSU) – Main Campus. The system is designed to manage and digitize scholarship information by efficiently storing, updating, and retrieving

scholarship records. It consolidates both internal and external scholarship programs within the higher education institution, including academic scholarships for college and university scholars, as well as the institutional scholarship program for BiPSU employees, known as the BiPSU Staff Scholarship.

The system also covers external scholarship programs offered by agencies such as the Department of Science and Technology (DOST), including the Science and Technology (S&T) Undergraduate Scholarships and the Junior Level Science Scholarship (JLSS); the Commission on Higher Education (CHED), including the CHED Merit Scholarship Program (CMSP) and the Coconut Scholarship Program (CoScho); and the Tertiary Education Subsidy (TES) and Tulong Dunong Program (TDP). It allows students and employees to view available scholarship opportunities along with their corresponding requirements.

Furthermore, the system processes student data to generate eligibility recommendations and rank qualified applicants using a Rule-Based Filtering approach. It also includes functionalities for screening student qualifications and generating reports that support institutional, accreditation, and international reporting requirements. The evaluation of the system will be guided by the ISO/IEC 25010 Software Quality Standards.

The system encompasses the essential functions of scholarship information management within the university, including scholarship record management, automated data retrieval, eligibility recommendation generation, and administrative report production. The eligibility recommender relies on



student-provided information, existing records from the SDSO Office and the UniFAST Office, and predefined scholarship qualification criteria to identify suitable scholarship programs.

However, this study is limited to scholarship data management, retrieval, and recommendation processes within BiPSU. The eligibility recommender is intended only for the Affirmative Scholarship Program, BiPSU Staff Scholarship, and Tertiary Education Program (TES) and does not cover other scholarship programs offered by the university. Furthermore, the system does not include the final approval or rejection of scholarship applications, fund disbursement, budget monitoring, or integration with external government agencies such as CHED and DOST. These functions remain under the responsibility of authorized personnel and separate institutional units.

Unlike the Student Information and Accounting System (SIAS), which manages enrollment, grades, and financial records, this system focuses exclusively on scholarship-related processes and does not support broader student information management functions.

The system is limited to a web-based platform and does not include mobile application development. Additionally, it excludes advanced artificial intelligence (AI) and predictive analytics features, with recommendation capabilities restricted to rule-based eligibility matching.

### **Definition of Terms**

As a reference for readers, the following are the technical terms and concepts used in this study:

*Back-end.* The back end refers to the server side of a software application or website. It is essentially the part of the system users don't see but handles the behind-the-scenes operations, such as managing data, processing requests, and ensuring everything runs smoothly.

*Rule-Based Filtering.* A recommendation technique that generates suggestions based on predefined rules, conditions, and eligibility criteria. In this study, it is used to evaluate student profiles such as academic performance, beneficiary status, employment status, and other qualifications against scholarship requirements to generate personalized eligibility recommendations.

*Framework.* A framework is like a pre-built structure or set of tools that helps developers build software applications more efficiently. It provides a foundation with ready-made components, libraries, and guidelines, saving developers time and effort.

*Front-end.* Front-end refers to the part of a website or application that users interact with directly. It encompasses everything users see, touch, click, or interact with, including the design, layout, content, and user interface.

*Recommender System.* A software component designed to generate personalized suggestions for users based on predefined criteria, user attributes, or similarity algorithms. In this study, the recommender system identifies and ranks scholarship programs according to a student's eligibility and suitability.

*Scholarship.* A financial assistance program awarded to students based on academic merit, financial need, or specific eligibility criteria set by institutions or government agencies. In this study, scholarships refer to the

internal, external, and national scholarship programs offered or implemented at Biliran Province State University, which provide educational support to qualified students.

*Scholarship Management System.* A digital platform that organizes, stores, and retrieves scholarship-related data. In this study, it serves as the centralized system for managing internal and external scholarship records, tracking applicant profiles, and generating administrative reports for Biliran Province State University.

*Web-Based System.* A software application accessed through a web browser over the internet or an internal network. In this study, it acts as the platform through which scholarship data are digitized, retrieved, updated, and evaluated, and where scholarship recommendations are generated.

*ISO/IEC 25010.* An international software quality standard that provides a framework for evaluating the quality of software products and systems. In this study, ISO/IEC 25010 serves as the basis for assessing the performance, usability, security, maintainability, and overall quality of the developed scholarship management and recommender system.

## **CHAPTER II**

### **RELATED LITERATURES**

This chapter presents the relevant literature and studies that are relevant and provide valuable insights into the development and implementation of the Scholarship Record Management with Eligibility Recommender for Higher Education Institution.

#### **Scholarship Program in the Philippines**

Scholarship programs play a crucial role in expanding educational access and supporting financially disadvantaged students in the Philippines. Various government-funded and institution-based scholarship initiatives have been developed to alleviate educational expenses and promote equitable participation in higher education. According to the Commission on Higher Education (CHED), scholarships such as the Tertiary Education Subsidy (TES), Merit Scholarship, and Tulong Dunong Program have been essential in reducing dropout rates, improving retention, and assisting low-income families in coping with rising educational costs. However, despite their benefits, scholarship systems across the country still face persistent administrative and logistical challenges that hinder efficient service delivery and affect student experiences.

Several studies highlight recurring concerns in scholarship management. A national evaluation of the TES program showed that administrative bottlenecks, slow agency coordination, and manual validation procedures lead to delays in eligibility checking and fund release (Bayudan-Dacuycuy, Orbeta, Vargas, and Ortiz, 2024). These delays reduce the

effectiveness of the subsidy and create financial pressure for students who rely on timely support. Likewise, Chao (2022) emphasized that although scholarships improve educational access, inefficiencies in processing and inconsistent documentation requirements prevent students from fully benefiting from these programs. These issues reflect a gap between scholarship policies and their implementation and demonstrate the need for more organized and technology-supported approaches.

At the institutional level, local studies further emphasize student experiences related to scholarship implementation. Ablian et al. (2023) found that while the Free Higher Education (FHE) law improved enrollment and financial relief among students, its retention policy implementation in local colleges faced issues such as inconsistent enforcement, confusing guidelines, and delayed notifications. These administrative inconsistencies often resulted in student disqualification, affecting scholarship continuity. Similar issues were reported by Cagasan, Belonias, and Cuadra (2020), whose study on scholarship recipients' experiences showed that while financial grants supported academic success, barriers such as long processing times, poor communication, and unclear procedures limited students' ability to maximize scholarship benefits.

Longitudinal research further underscores the importance of proper record management and ongoing monitoring in scholarship programs. Aguaras (2025), who examined retention and graduation patterns of Iskolar ng Bayan scholars, revealed that efficient data management systems, timely communication, and accurate monitoring strongly influence scholars' long-term academic performance and completion rates. The study emphasized that

scholarship success depends not only on financial support but also on how effectively scholarship offices manage records, monitor academic performance, and communicate requirements.

More recent evaluations of student perceptions also highlight gaps in service delivery. General et al. (2025) found that while TES grantees reported high satisfaction with the support provided by scholarship personnel, delays in fund disbursement and documentation inconsistencies remained significant concerns. These operational challenges reinforce the need for increased coordination among agencies and the adoption of digital tools to improve scholarship workflow efficiency, transparency, and responsiveness.

### **Scholarship Application Process in the Philippines**

Scholarship applications in the Philippines typically involve structured procedures requiring academic records, income-related documents, and formal application submissions. Despite this structure, manual and paper-based processing remains common, resulting in delays, misplaced documents, and slow verification. These administrative inefficiencies become more pronounced as the demand for scholarships increases and as agencies struggle to handle large volumes of applicants using traditional workflows. Research shows that manual processes in local scholarship offices often create operational burdens and compromise the timely delivery of services, emphasizing the need for more organized and technologically supported systems (Ablen, Flores, Pasturan, & Bertulfo, 2025).

National-level evaluations further highlight the systemic issues in scholarship administration. In their comprehensive assessment of the Tertiary

Education Subsidy (TES) program, Bayudan-Dacuycuy, Orbeta, Vargas, and Ortiz (2024) found that bottlenecks caused by slow inter-agency coordination and manual validation procedures delay eligibility checking and fund release. Morato (2022) also stressed that although scholarships significantly improve access to higher education for financially disadvantaged students, their positive impact is undermined when administrative delays and inconsistencies prevent timely approval and disbursement. These findings collectively demonstrate the need for streamlined and systematic approaches to scholarship management to ensure fairness, timeliness, and effective program implementation.

Recent studies strongly recommend digital transformation to address these challenges in scholarship processing. Digitized scholarship systems have been shown to improve accuracy, enhance transparency, and reduce processing time through automated workflows, centralized records, and efficient communication channels (Nagrama, Teñajura, Taer, & Timod, 2025). Research on local scholarship programs further highlights that students express greater satisfaction and motivation when scholarship procedures are clearly communicated, well-organized, and supported by reliable information systems (Lacorte, Espina, Prado, & Lerin, 2025). Additionally, the integration of data-driven and predictive tools, such as machine learning models for scholarship eligibility, can significantly enhance the fairness and efficiency of applicant screening (Fajardo et al., 2024). These developments support the creation of a College Scholarship Record Management and Retrieval System with an Eligibility Recommender, which aligns with the documented need for

more efficient, accurate, and transparent scholarship administration in the Philippines.

### **Scholarship Recommendation Systems**

Scholarship recommendation systems have become important tools in assisting students in identifying scholarship opportunities that match their qualifications, academic performance, and personal background. Traditional scholarship selection and searching processes are often difficult and time-consuming due to the large number of scholarship programs and varying eligibility requirements. Recommendation systems help automate the matching process and provide personalized suggestions to students.

Anggrawan et al. (2022) developed a scholarship recipients recommendation system using the Analytical Hierarchy Process (AHP) and MOORA methods. Their study aimed to improve the scholarship selection process by evaluating applicants based on multiple criteria such as academic performance and socioeconomic factors. The researchers found that recommendation systems can improve decision-making accuracy and reduce bias in scholarship allocation.

Similarly, Fajardo et al. (2024) explored the use of machine learning in predicting scholarship grants in a local government unit in the Philippines. Their data-driven approach demonstrated that predictive analytics can effectively identify deserving students and improve scholarship management processes. The study emphasized the importance of intelligent systems in increasing accessibility and efficiency in scholarship programs.

The study of Alfonso et. al. (2024), entitled “Mediskolar: a Web-Based



Scholarship Management System with Profile Analysis Using Analytical Hierarchy Process (AHP) and Decision Tree Algorithm for Marikina City” discussed the development of a web-based scholarship management system that utilized AHP and Decision Tree algorithms to analyze student profiles and scholarship eligibility. The study highlighted the effectiveness of integrating intelligent decision-support methods in scholarship management systems to improve applicant evaluation and recommendation processes.

Sunny and C (2025) reviewed existing scholarship recommendation systems and identified several gaps, including lack of personalization and limited recommendation accuracy. The authors suggested that intelligent recommendation approaches, such as content-based and hybrid filtering methods, can improve scholarship recommendation systems by providing more accurate and user-centered recommendations.

### **Rule-Based Filtering in Recommendation Systems**

Rule-Based Filtering is a recommendation technique that generates suggestions based on predefined rules, conditions, and domain knowledge. Unlike similarity-based recommendation approaches, rule-based filtering evaluates user characteristics against established criteria to determine appropriate recommendations. In scholarship recommendation systems, this approach can match student profiles with scholarship eligibility requirements to identify suitable opportunities.

Engin et al. (2014), in their study entitled "Rule-Based Expert Systems for Supporting University Students," developed a rule-based expert system to assist university students in making academic decisions. The system utilized

predefined rules derived from expert knowledge to provide recommendations based on students' specific situations and needs. The findings demonstrated that rule-based systems can effectively assist students by delivering accurate and consistent recommendations. This suggests that rule-based filtering can be applied in scholarship recommendation systems where recommendations must comply with predefined eligibility requirements.

Similarly, Ketter et al. (2008), in their study "A Semantic Web Architecture for Advocate Agents to Determine Preferences and Facilitate Decision Making," discussed knowledge-based recommender systems that utilize domain knowledge and predefined rules to generate recommendations. The researchers explained that rule-based recommendation techniques are particularly effective when user requirements must be matched against explicit constraints and criteria. Their study highlighted that such systems improve decision-making by ensuring that all relevant conditions are satisfied before recommendations are presented. This concept is applicable to scholarship recommendation systems because scholarship qualifications are often based on clearly defined academic, financial, and demographic criteria.

Urdaneta-Ponte et al. (2021), also emphasized in their study "Recommendation Systems for Education: Systematic Review," examined various recommendation approaches used in educational environments. The review found that knowledge-based and rule-based recommendation systems are widely implemented when user characteristics, qualifications, and eligibility requirements are explicitly defined. The authors emphasized that rule-based systems provide transparent and explainable recommendations, allowing users to understand the basis of each recommendation. This feature

is particularly valuable in scholarship recommendation systems, where students need clear justification regarding their eligibility for scholarship opportunities.

### **Intelligent and Hybrid Recommendation Systems**

Modern recommendation systems increasingly integrate intelligent and hybrid techniques to improve recommendation quality and accuracy. Hybrid recommendation systems combine multiple recommendation methods to overcome the limitations of individual approaches

Hazar (2025) introduced an intelligent hybrid recommendation system for e-learning personalization in smart campuses. The system combined content-based filtering with intelligent algorithms to provide personalized learning recommendations. The findings revealed that hybrid recommendation techniques improved recommendation accuracy and user satisfaction. Although focused on e-learning, the study is relevant because scholarship recommendation systems also require personalized matching based on user profiles and preferences.

Zhang et al. (2023) conducted a literature survey on scholarly recommendation systems and highlighted that content-based filtering remains one of the most reliable methods for personalized recommendations. The researchers explained that recommendation systems can be enhanced further through machine learning and deep learning techniques.

Stergiopoulos et al. (2024) proposed an academic recommender system based on clustering, graph modeling, and deep learning. Their study demonstrated that advanced recommendation techniques can efficiently

process large datasets and generate highly relevant recommendations. The study supports the integration of intelligent technologies into educational recommendation platforms, including scholarship recommendation systems.

### **Fairness in Recommendation Systems**

Fairness is an important consideration in recommendation systems because biased recommendations may disadvantage certain users or groups. In scholarship recommendation systems, fairness ensures that all students receive equal opportunities to access suitable scholarship recommendations based on their qualifications.

Wang et al. (2022) conducted a comprehensive survey on fairness in recommender systems and discussed various sources of bias that may affect recommendation outcomes. The study emphasized that fairness-aware recommendation systems should provide balanced and unbiased recommendations while maintaining recommendation accuracy. The researchers highlighted the importance of transparency, equitable data representation, and ethical recommendation practices. This concept is relevant to scholarship recommendation systems because fair recommendations can help ensure equal access to scholarship opportunities for students from diverse backgrounds.

## CHAPTER III

### DESIGN AND METHODOLOGY

This chapter provides an overview of the methodology, requirement analysis, requirement documentation, and design of software, systems, products, and processes.

#### **Methodology**

The proponents utilized the Agile methodology, following the principles of the conventional Software Development Life Cycle (SDLC). Agile is characterized by iterative development, continuous collaboration, and adaptability to changes in user requirements. Under this approach, the project was divided into multiple incremental phases, and regular consultations with target users ensured accurate alignment with the scholarship office's operational needs. Figure 3.1 illustrates the Agile Model applied to this study.

The fundamental principle of Agile is establishing the intended users' needs, the scope of the problem, the opportunities for improvement, and the value the system must deliver. Agile emphasizes breaking the project into smaller, manageable components and encourages continuous improvement throughout development (GeeksforGeeks, 2026).

This methodology is well-suited to the study due to the complexity of scholarship processes, ongoing updates to scholarship programs, and the dynamic nature of user requirements. Through iterative design and testing, key system features including scholarship record management, automated data retrieval, and scholarship eligibility recommendation were continuously refined based on user feedback.

The method employed in creating this web application followed the Agile ADLC Model as shown in figure 3.1.

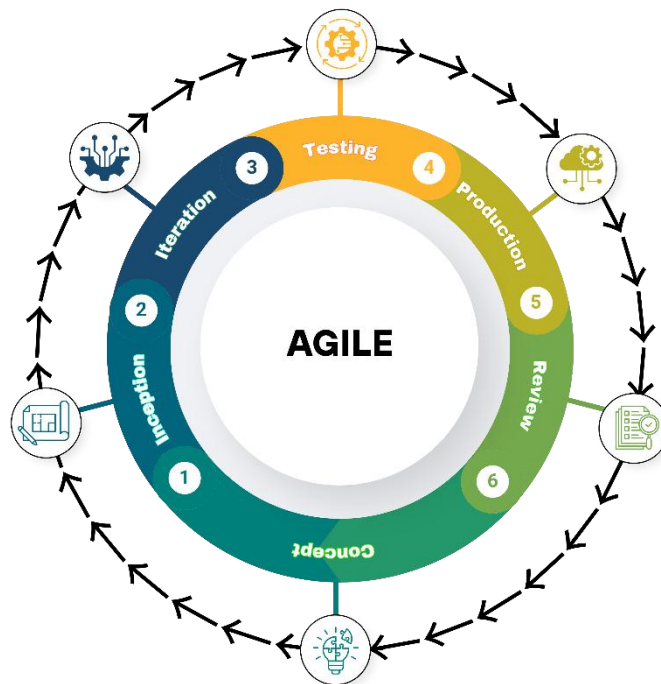


Figure 3.1. Agile ADLC Model

## Requirements Engineering

To gain a comprehensive understanding of the existing scholarship process, the proponents conducted interviews with the Scholarship Coordinator, Focal Person and Dean or any representative from the Student Development and Services Offices (SDSO) Staff Office and UniFAST Office of Biliran Province State University (BiPSU) Main Campus. The questions used in the interviews were patterned after the PIECES Framework (Performance, Information, Economy, Control, Efficiency, Services), allowing the identification of operational challenges in the current manual scholarship workflow.

After determining the problems, the proponents were able to define the functional and non-functional objectives of the proposed web-based solution.

#### *Functional Requirements*

1. The system shall allow students to submit scholarship requirements online, which will reduce manual submission and improve the efficiency of document collection in the Student Development and Services Office (SDSO) and UNIFAST Office.
2. The system shall store and manage scholarship records in a centralized and secure database, which can be used to organize and maintain accurate and accessible records of scholars.
3. The system shall provide fast and accurate retrieval of scholarship records using Rule-Based Filtering, which will help staff quickly access needed information and reduce time spent searching for files.
4. The system shall generate reports automatically, which can assist in timely submission of required reports to the for SDSO and UniFAST Office.
5. The system shall provide data analytics features that present scholarship data through summaries and visualizations, which can be used to monitor scholar status and support decision-making.
6. The system shall allow authorized users to access the platform based on their roles, ensuring proper management and control of scholarship records.

#### *Non-Functional Requirements*

1. **Functionality.** The system shall be accessible through common web browsers such as Google Chrome, Mozilla Firefox, and Safari to ensure usability across different platforms.
2. **Reliability.** The system shall ensure accurate storage, retrieval, and processing of scholarship records, and maintain consistent performance with minimal errors during operations.
3. **Usability.** The system shall provide a user-friendly interface for both students and staff, with clear navigation and easy-to-understand features for managing scholarship records.
4. **Efficiency.** The system shall minimize processing time in searching, retrieving, generating reports and efficient data handling techniques.
5. **Maintainability.** The system shall be designed to allow easy updates, modifications, and improvements to accommodate future enhancements and changes in requirements.
6. **Portability.** The system shall be deployable and accessible across different devices and environments without requiring major adjustments.
7. **Compatibility.** The system shall function properly when accessed by multiple users using different devices, operating systems, and browsers.
8. **Security.** The system shall implement authentication and authorization mechanisms, such as password protection and user roles, to ensure the confidentiality and integrity of scholarship records and sensitive data.



## Design of Software, System Product, or Process

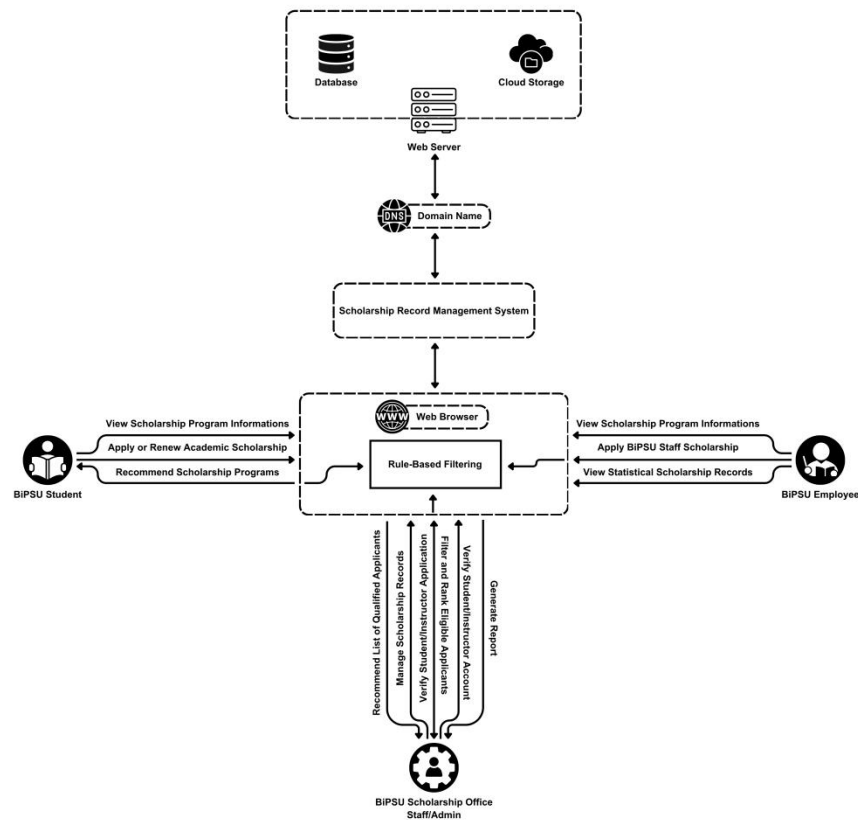


Figure 3.2. **System Architecture**

Once the requirements were established, the next stage focused on the design of the system's behavior, structure, and operational flow. Figure 3.2 presents an overview of the system architecture. The proposed system utilizes a cloud-based architecture to store and manage scholarship data efficiently. Centralized storage ensures that records from applicant information to scholarship details can be accessed securely by authorized users anytime and anywhere.

Upon submitting their scholarship applications, student data and attached documents are automatically organized within the system. This enables streamlined access, automated retrieval, and efficient processing. The eligibility recommender analyzes student information and compares it to

scholarship criteria, generating personalized scholarship recommendations based on similarity scores.

The system design also ensures that scholarship staff can manage records, generate reports, validate eligibility, and monitor scholarship statuses with minimal manual effort.

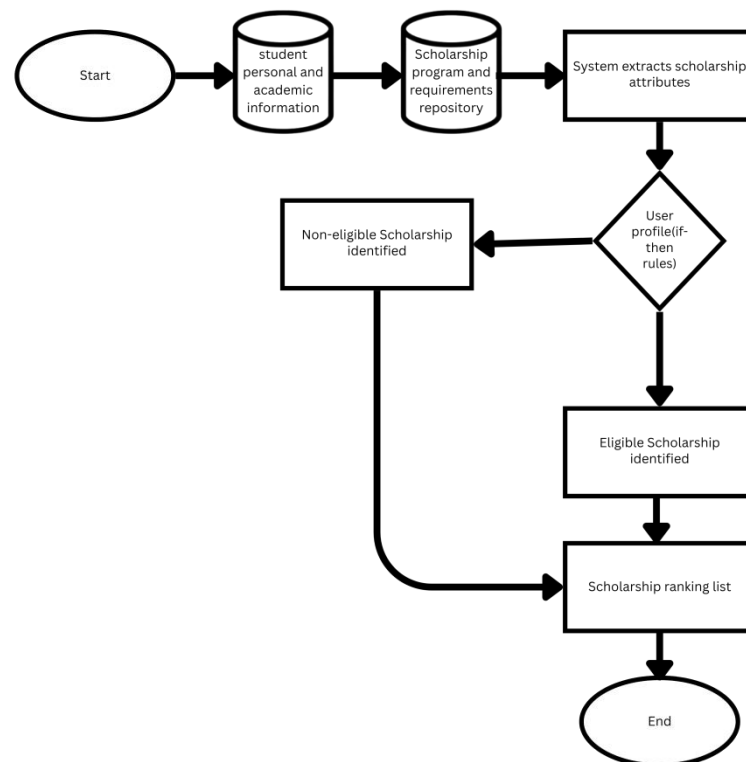


Figure 3.3. **Rule Based Filtering Process Flow**

In Figure 3.3, the algorithm applied for the recommender is rule-based filtering, which uses personal preferences and eligibility information to recommend scholarship programs to BIPSU students. The key necessary components for recommending a scholarship program include the unique eligibility attributes, criteria, and qualification indicators of each scholarship program. After that, the system retrieves the scholarship attributes from the

repository and undergoes a rule evaluation process, where the student's personal, academic, and eligibility information is systematically compared against the requirements of each scholarship program using predefined if-then rules. Based on the evaluation results, the system identifies both eligible and non-eligible scholarships according to whether the student satisfies the specified qualifications. Lastly, the resulting evaluation outcomes are processed to generate a scholarship ranking list, which prioritizes scholarship opportunities based on the degree of compatibility between the student's profile and the scholarship requirements, ensuring that the most suitable scholarship programs are presented to the student according to their overall qualifications and eligibility status.

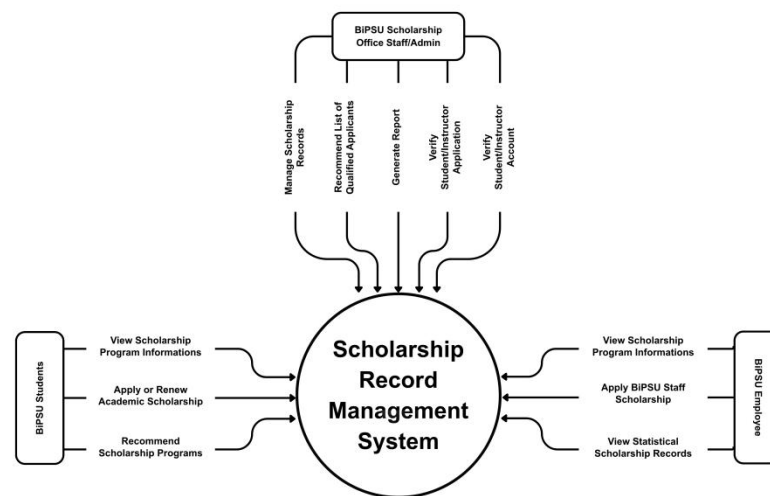


Figure 3.4. **Context Diagram**

Figure 3.4 below display the central system and its main stakeholders. The SDSO/UNIFAST Staff utilizes the system to generate reports, encode student requirements, and update student requirements. BIPSU students interacts with the system by signing in or signing up, applying for a scholarship program, and receiving a recommended list of scholarship programs. The Admin engages with the system to generate reports, verify

student applicants, and upload scholarship requirements/ Super Admin holds the highest level of system access, responsible for generating reports, updating user accounts, and managing school personnel and admin account creation to maintain the overall integrity and security of the system. This system combines these interactions into a centralized platform that streamlines the scholarship application and management process

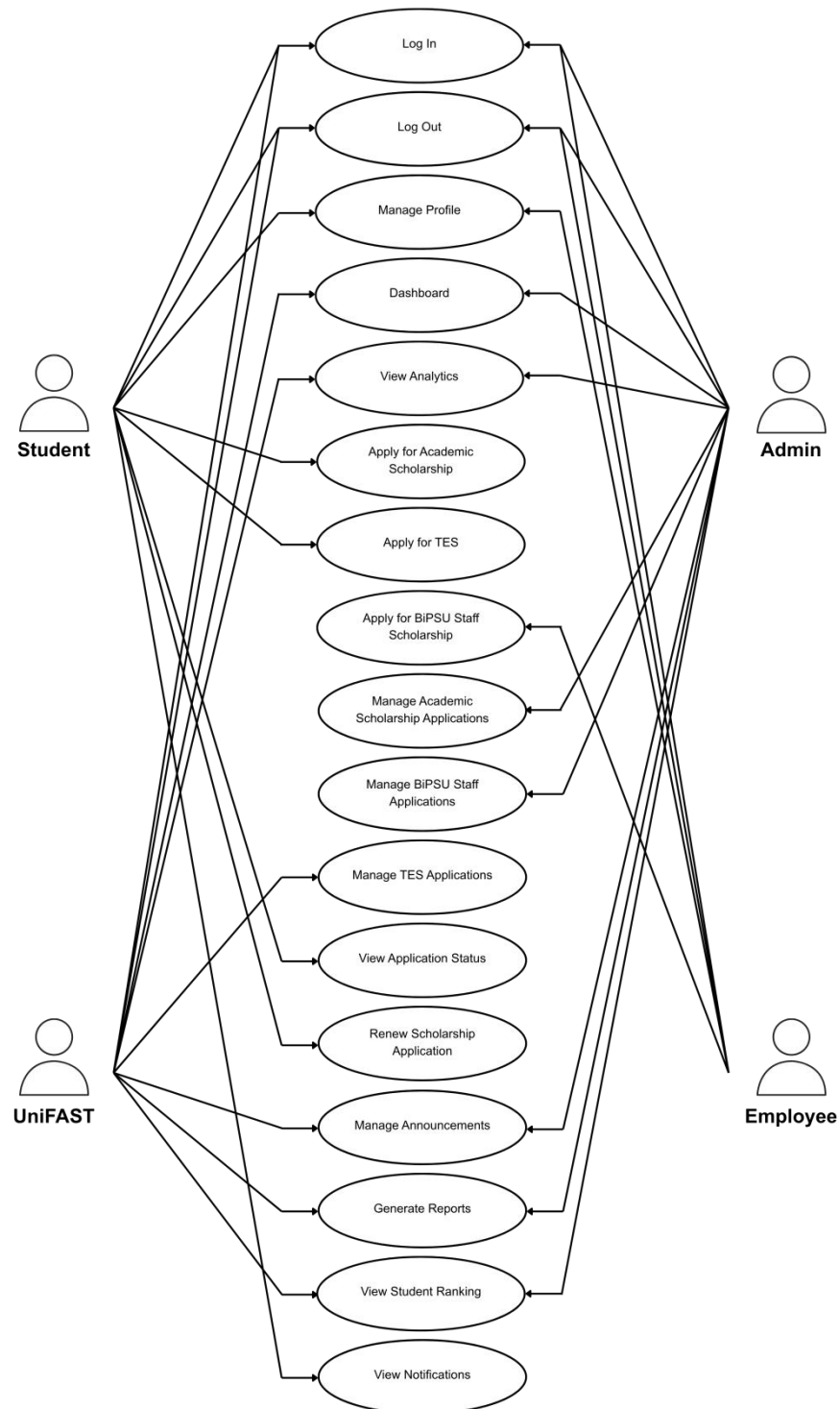


Figure 3.5. Use Case Role

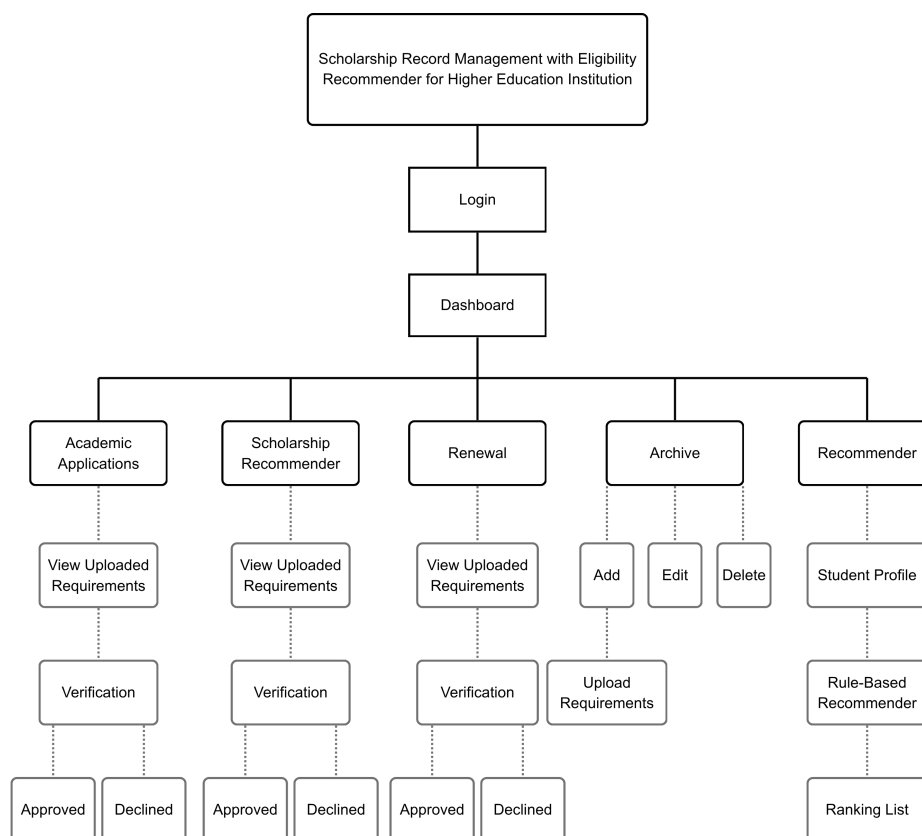


Figure 3.6. Site Map for Admin

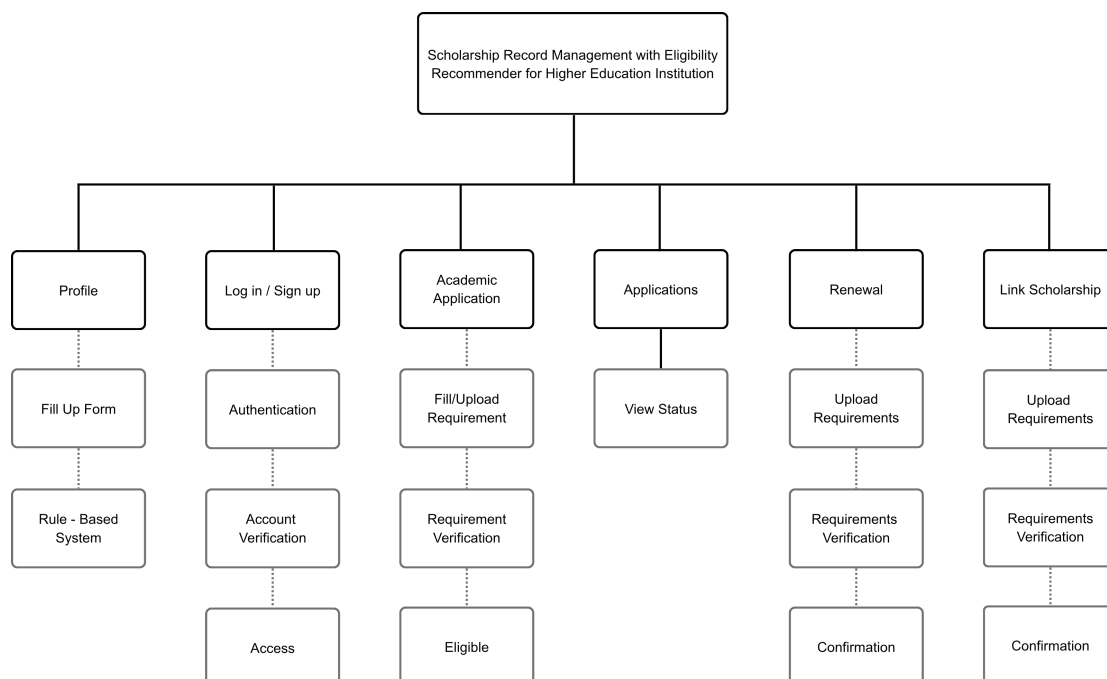


Figure 3.7. Site Map for Student

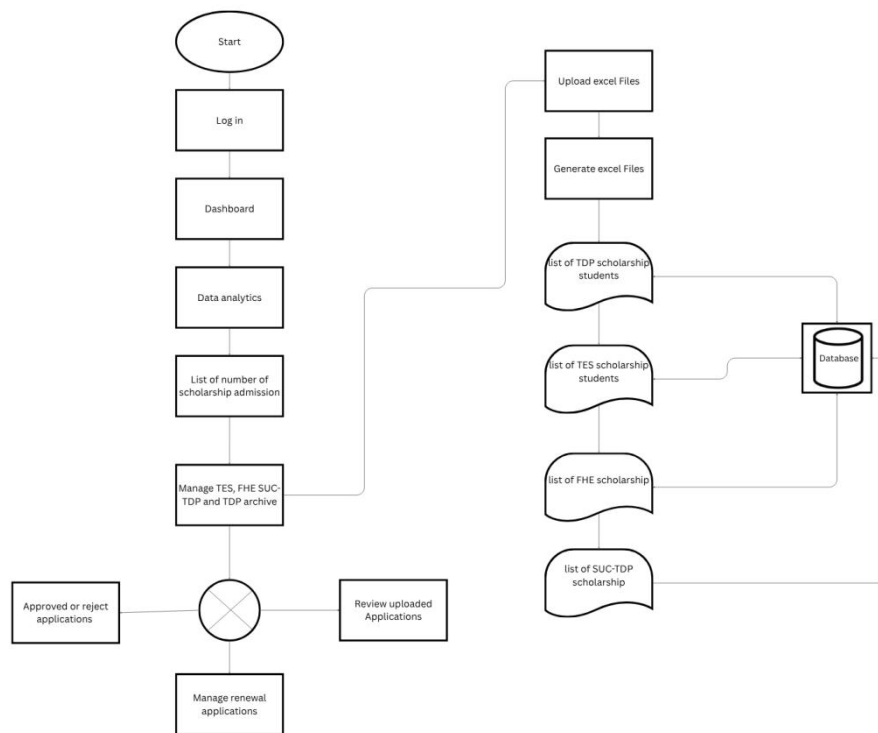


Figure 3.8. **System Flowchart for UNIFAST Office**

Figure 3.6 present the system flowchart illustrates the sequential process for managing scholarship applications and records within the UNIFAST Office system. The user must log in to gain access to the system, which then directs them to the dashboard, where they can navigate to the data analytics module to monitor and review relevant scholarship data. From there, the system provides a list of the number of scholarship admissions, allowing administrators to manage the TES, FHE, and TDF archives accordingly. The system also supports the uploading and generation of Excel files, which are used to populate and maintain the database containing the list of TDP scholarship students, TES scholarship students, and FHE scholarship students, all of which are interconnected and synchronized with the central database. Furthermore, administrators are able to review uploaded applications through a decision point in the system, where they can either approve or reject applications based on the evaluation of submitted

documents. Additionally, the system accommodates the management of renewal applications, ensuring that continuing scholars are properly processed and updated.

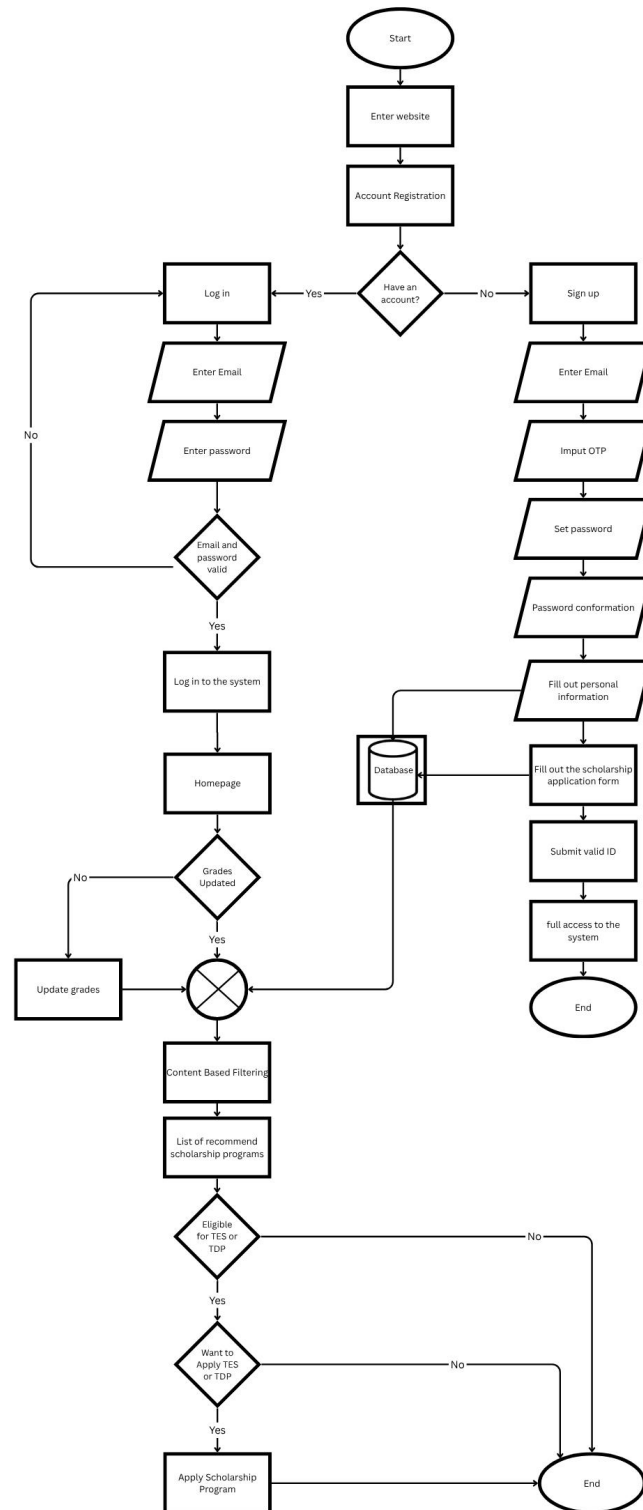


Figure 3.9. **System Flowchart for BIPSU Student's Applicant**



The diagram shown in Figure 3.7 shows the college applicants system flow, including the sequential procedure of account registration via the process of signing up, ensuring that all transactions are kept in the database and the applicants must provide their personal information and upload the essential requirements to facilitate the curation of recommended scholarship programs. Once the registration is complete and the user can log in to the system. Following the completion of the account registration, applicants can view the recommended scholarship programs. The final selection of a scholarship program will be determined by the applicants themselves.

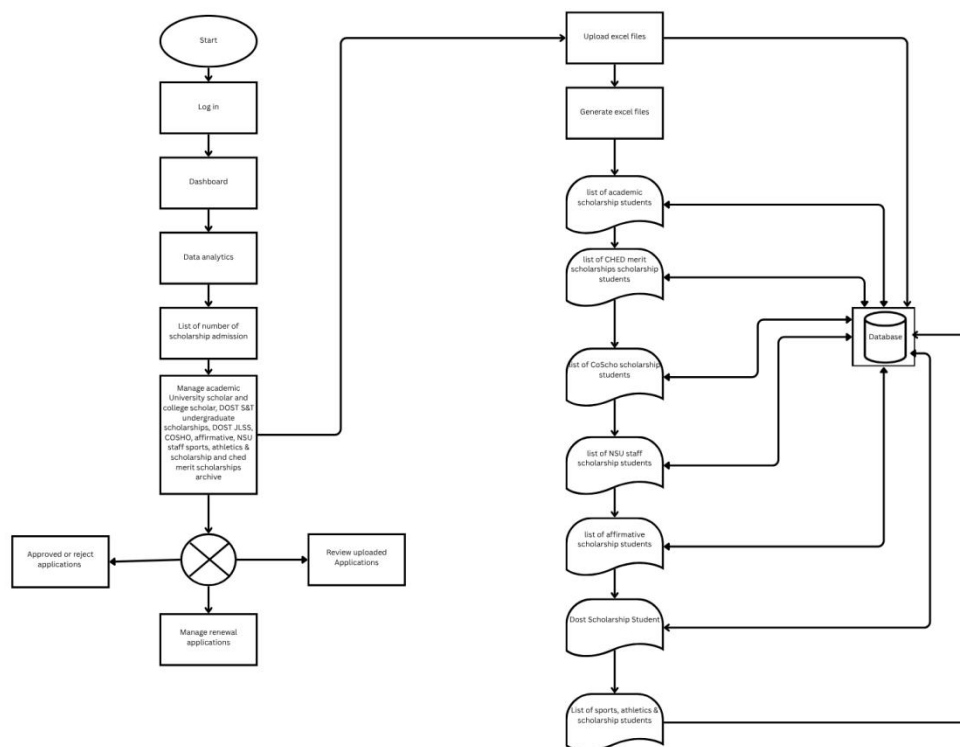


Figure 3.10. **System Flowchart for SDSO Office**

In figure 3.8 the system flowchart illustrates the sequential process for managing and monitoring various scholarship programs within the SDSO Office system. Initially, the user must log in to gain access to the system,

which directs them to the dashboard where they can navigate to the data analytics module to oversee and analyze scholarship-related data and statistics. From there, the system presents a list of the number of scholarship admissions, enabling administrators to manage students under multiple scholarship categories, including university scholars, college scholars, DOST, SKI undergraduate scholarships, SGD, staff sports, athletes, scholarship and child welfare programs, and government service. The system also supports the uploading and generation of Excel files, which serve as the primary means of populating and synchronizing the central Database that holds records of academic scholarship students, CHED-merit scholarship students, Valiente scholarship students, NGO staff scholarship students, affirmative scholarship students, First Scholarship students, and sports and athletics scholarship students, all of which are consistently interconnected with the database to ensure accurate and up-to-date record keeping across all scholarship classifications. Furthermore, administrators are able to review uploaded applications through a critical decision point in the system, where they can either approve or reject applications based on a thorough evaluation of the submitted requirements and qualifications. The system also accommodates the efficient management of renewal applications, ensuring that continuing scholars across all scholarship categories are properly processed, monitored, and updated in a well-structured and organized manner.

## Database Schema

Figure 3.11. Entity Relationship Diagram (ERD)

Table 3.0. User (Example)

| Column Name | Data Type | Attributes | Description | Length |
|-------------|-----------|------------|-------------|--------|
|             |           |            |             |        |
|             |           |            |             |        |
|             |           |            |             |        |
|             |           |            |             |        |
|             |           |            |             |        |

## User Interface Mock-ups

To provide a comprehensive perspective of the system interface, mock-up designs were created throughout the design ph. These mock-ups serve as visual representations of the system and are used to assist in the development, testing, and feedback gathering from users.

The system will provide an online admission application form for college applicants, shown below in Figure 3.13 to easily access and send applications to the admission office. The information contained in the form was based on the current admission application form provided by the admission office. It includes the personal details of applicants.

Figure 3.12. **Landing Page**

Figure 3.5. **Student Scholarship Applicants/Student HIPO Data Flow Diagram**

Figure 3.5. **Faculty Scholarship Applicants HIPO Data Flow Diagram**

Figure 3.5. **SDSO Staff HIPO Data Flow Diagram**

Figure 3.5. **UniFAST Staff HIPO Data Flow Diagram**

## **Development Plan**

To manage the development of the system effectively, the Gantt chart serves as a visual reference to outline the project's timeline, allocate resources, and track the completion of tasks across various phases. It will be shown in Figure 3.9 that starts in planning, data collection and preprocessing, model development, design process, front-end development, back-end development, the model integration into the application, testing, and validation, deployment, and lastly is the post-launch monitoring. It is estimated that the project will be finished in two semesters only.

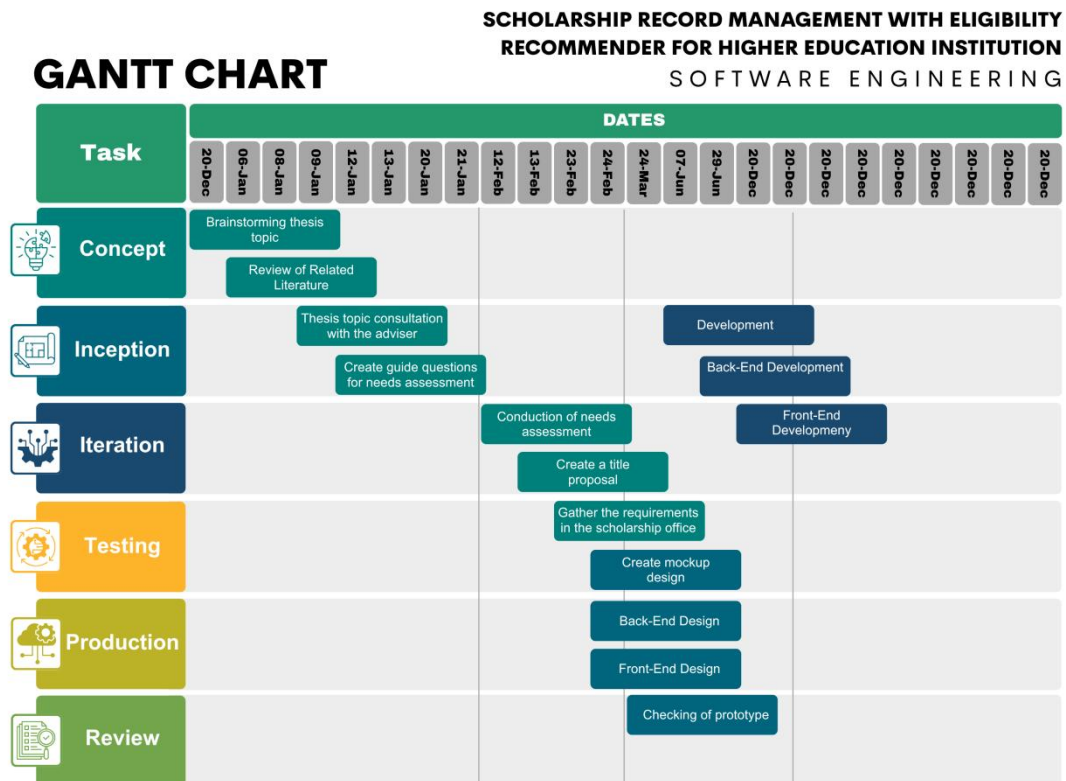


Figure 3.13. Gantt Chart

## Technical Background

Developing the College Scholarship Record Management and Retrieval System with Scholarship Program Eligibility Recommender requires the integration of several modern technologies, tools, and development frameworks that support reliable data processing, efficient retrieval, and intelligent recommendation capabilities. As a web-based system, the project utilizes standard front-end technologies such as HTML, CSS, and JavaScript to design user interfaces that are responsive, accessible, and easy to navigate across devices and browsers. These technologies ensure a consistent user experience for students and administrators interacting with scholarship records and eligibility results, aligning with best practices in web usability (Learn Web Development | MDN, 2025).

For server-side development, the system may employ backend technologies such as PHP, Python, or Node.js, which handle core operations including data processing, user authentication, and secure communication between the client interface and the database. Backend frameworks allow the system to process large sets of scholarship data efficiently, manage user roles, and perform automated eligibility checking based on predefined criteria.

To support secure and structured data storage, the project integrates a Database Management System (DBMS) such as MySQL, MariaDB, PostgreSQL, or Firebase. These relational and cloud-based databases allow efficient indexing, high-speed retrieval, and accurate handling of scholarship-related records including student information, requirements, compliance, status tracking, and historical data. As noted by Jones (2024), the use of relational databases ensures data integrity and supports complex queries necessary for scholarship verification and reporting.

A key component of the system is the Eligibility Recommender, which relies on machine learning and similarity-based techniques. Tools such as Scikit-learn, Pandas, or NumPy in Python may be used to implement Content-Based Filtering supported by Cosine Similarity, enabling the system to analyze student profiles and match them with scholarship programs based on academic performance, socioeconomic background, and program requirements. These libraries provide robust algorithms for classification, data preprocessing, and prediction automating the eligibility evaluation process and minimizing subjective biases in decision-making.

To facilitate system development, testing, and debugging, the project utilizes integrated development environments (IDEs) such as Visual Studio Code, Sublime Text, or Jupyter Notebook. These development tools enhance productivity by offering syntax support, code refactoring, version control integration, and visualization of machine learning outputs. They also help ensure that the system is scalable, maintainable, and adaptable to future enhancements.

Modern web development practices may also incorporate version control platforms such as Git and GitHub for collaborative coding, tracking changes, and ensuring code stability during iterative development cycles. Meanwhile, testing tools and browser-based debugging utilities verify system performance, data validation, and responsiveness prior to deployment.

## Testing Plan

The proposed Scholarship Record Management System (SRMS) with Eligibility Recommender evaluated by various user including, the (4) admission staff, (9) college students, (2) school personnel and (2) ICT experts using the ISO 25010, an internationally recognized standard for software evaluation. The nine primary quality standards are defined as follows: 1) Functional Suitability, 2) Performance Efficiency, 3) Compatibility, 4) Interaction Capability, 5) Reliability, 6) Security, 7) Maintainability, 8) Flexibility, and 9) Safety, with a rating of 5 for very high extent, 4 for high extent, 3 for moderate extent, 2 for low extent, and 1 for very low extent.

Table 3.15. **Descriptive Interpretation of Evaluation Process**

| <b>Mean Score</b> | <b>Descriptive Interpretation</b> |
|-------------------|-----------------------------------|
| 4.21 – 5.0        | Very High Extent                  |
| 3.41 – 4.20       | High Extent                       |
| 2.61 – 3.40       | Moderate Extent                   |
| 1.81 – 2.60       | Low Extent                        |
| 1.0 – 1.80        | Very Low Extent                   |

## **Implementation Plan**



## CHAPTER IV

### DESIGN AND METHODOLOGY

This section presents the results and corresponding discussion of the study.

**Developed a web-based application**

**Integrate a scholarship eligibility recommender**

**Report Generation Module**

**Evaluate the developed software based on ISO 25010 or International Software Quality Standards**

Table 4.1. **Functional Suitability Evaluation Results**

| Indicators                               | Mean | Description |
|------------------------------------------|------|-------------|
| 1. Functional completeness               |      |             |
| 2. Functional correctness                |      |             |
| 3. Functional appropriateness            |      |             |
| <b>Total Average Weighted Mean (AWM)</b> |      |             |

Table 4.2. **Performance Efficiency Evaluation Results**

| Indicators                               | Mean | Description |
|------------------------------------------|------|-------------|
| 1. Time behaviour                        |      |             |
| 2. Resource utilization                  |      |             |
| 3. Capacity                              |      |             |
| <b>Total Average Weighted Mean (AWM)</b> |      |             |

Table 4.3. **Compatibility Evaluation Results**

| Indicators | Mean | Description |
|------------|------|-------------|
|------------|------|-------------|

|                                          |
|------------------------------------------|
| 1. Co-existence                          |
| 2. Interoperability                      |
| <b>Total Average Weighted Mean (AWM)</b> |

Table 4.4. **Interaction Capability Evaluation Results**

| Indicators                               | Mean | Description |
|------------------------------------------|------|-------------|
| 1. Appropriateness recognizability       |      |             |
| 2. Learnability                          |      |             |
| 3. Operability                           |      |             |
| 4. User error protection                 |      |             |
| 5. User engagement                       |      |             |
| 6. User assistance                       |      |             |
| 7. Inclusivity                           |      |             |
| 8. Self-descriptiveness                  |      |             |
| <b>Total Average Weighted Mean (AWM)</b> |      |             |

Table 4.5. **Reliability Evaluation Results**

| Indicators                               | Mean | Description |
|------------------------------------------|------|-------------|
| 1. Faultlessness                         |      |             |
| 2. Availability                          |      |             |
| 3. Fault tolerance                       |      |             |
| 4. Recoverability                        |      |             |
| <b>Total Average Weighted Mean (AWM)</b> |      |             |

Table 4.6. **Security Capability Evaluation Results**

| Indicators         | Mean | Description |
|--------------------|------|-------------|
| 1. Confidentiality |      |             |
| 2. Integrity       |      |             |

3. Non-repudiation
4. Accountability
5. Authenticity
6. Resistance

---

**Total Average Weighted Mean (AWM)**

---

Table 4.7. **Maintainability Evaluation Results**

| Indicators                               | Mean | Description |
|------------------------------------------|------|-------------|
| 1. Modularity                            |      |             |
| 2. Reusability                           |      |             |
| 3. Analysability                         |      |             |
| 4. Testability                           |      |             |
| <b>Total Average Weighted Mean (AWM)</b> |      |             |

Table 4.8. **Flexibility Evaluation Results**

| Indicators                               | Mean | Description |
|------------------------------------------|------|-------------|
| 1. Adaptability                          |      |             |
| 2. Scalability                           |      |             |
| 3. Installability                        |      |             |
| 4. Replaceability                        |      |             |
| <b>Total Average Weighted Mean (AWM)</b> |      |             |

Table 4.9. **Safety Evaluation Results**

| Indicators                | Mean | Description |
|---------------------------|------|-------------|
| 1. Operational constraint |      |             |
| 2. Risk identification    |      |             |
| 3. Fall safe              |      |             |
| 4. Hazard warning         |      |             |

5. Safe integration

---

**Total Average Weighted Mean (AWM)**

---

Table 4.10. **Summary of Software Evaluation**

| Indicators                       | Mean | Description |
|----------------------------------|------|-------------|
| 1. Functional suitability        |      |             |
| 2. Performance efficiency        |      |             |
| 3. Compatibility                 |      |             |
| 4. Interaction-capability        |      |             |
| 5. Reliability                   |      |             |
| 6. Security                      |      |             |
| 7. Maintainability               |      |             |
| 8. Flexibility                   |      |             |
| 9. Safety                        |      |             |
| <b>Overall Evaluation Rating</b> |      |             |

## **CHAPTER V**

### **CONCLUSION AND RECOMMENDATION**

This chapter presents the summary of findings and recommendations to provide a comprehensive overview of the key findings and suggestions for future actions. These findings are relevant to the objective of the project to develop a scholarship record management system with eligibility recommender for higher education institution, as well as recommend programs for aspiring college applicants. The developed system was evaluated by target users as well as various experts, ensuring that the proposed solution is effective and beneficial.

#### **Conclusion**

#### **Recommendation**

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## **APPENDICES**

## Appendix A

### LETTER OF REQUEST TO CHOOSE A THESIS ADVISER



Republic of the Philippines

**BILIRAN PROVINCE STATE UNIVERSITY**

ISO 9001:2015 CERTIFIED

**SCHOOL OF TECHNOLOGY AND COMPUTER STUDIES**

January 8, 2026

**TITO AMERIGO V. CUSTODIO JR.**

ITE Faculty

BiPSU-STCS

Dear Sir:

Good day!

We, the undersigned, are third-year students of the Biliran Province State University pursuing the degree in **BS in COMPUTER SCIENCE** and are currently enrolled in **CS-3218 THESIS 1**.

We are writing to humbly request that you serve as our thesis adviser. We believe your knowledge and insights would be invaluable and greatly enrich our work.

Thank you for your consideration. We hope you will be able to fulfill our request.

Respectfully yours,

**ARIAN JAY G. MAALA**

**CARLVIN JAY G. SULLA**

**JUNIEL M. VILLABLANCA**

**SARAH JANE E. VILLANUEVA**

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\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

Noted by:

**REYMON M. SANTIAÑEZ, DIT**

Thesis 1 Instructor

Conforme:

**TITO AMERIGO V. CUSTODIO JR., MIT**

Thesis Adviser

## Appendix A

### LETTER OF REQUEST TO CHOOSE A THESIS CO-ADVISER



Republic of the Philippines

**BILIRAN PROVINCE STATE UNIVERSITY**

ISO 9001:2015 CERTIFIED

**SCHOOL OF TECHNOLOGY AND COMPUTER STUDIES**

August 5, 2026

**MARICEL S. SAULAN**

Scholarship In-Charge

Student Development and Services Office

Biliran Province State University

Dear Ma'am:

Good day!

We, the undersigned, are fourth-year students of the Biliran Province State University pursuing the degree in **BS in COMPUTER SCIENCE** and are currently enrolled in **CS-4123 THESIS 2**.

We are writing to humbly request that you serve as our thesis co-adviser. We believe your knowledge and insights would be invaluable and greatly enrich our work.

Thank you for your consideration. We hope you will be able to fulfill our request.

Respectfully yours,

**ARIAN JAY G. MAALA**

**CARLVIN JAY G. SULLA**

**JUNIEL M. VILLABLANCA**

**SARAH JANE E. VILLANUEVA**

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\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

Noted by:

**REYMON M. SANTIAÑEZ, DIT**  
Thesis 2 Instructor

**TITO AMERIGO V. CUSTODIO JR., MIT**  
Thesis Adviser

Conforme:

**MARICEL S. SAULAN**

Thesis Co-Adviser

## Appendix B

### LETTER OF REQUEST TO CONDUCT NEEDS ASSESSMENT



Republic of the Philippines

**BILIRAN PROVINCE STATE UNIVERSITY**

ISO 9001:2015 CERTIFIED

**SCHOOL OF TECHNOLOGY AND COMPUTER STUDIES**

January 8, 2026

**GRETCHEN STEPHANIE M. ESPINA**

Municipal Mayor

Municipality of Naval

Brgy. P.I. Garcia, Naval, Biliran

Dear Ma'am:

In this regard, we respectfully request approval to conduct a **NEEDS ASSESSMENT** in the community of Naval and Higatangan Island, including interviews with your office and constituents to identify key needs and issues for our project proposal and thesis topic selection.

Rest assured that all information obtained will be treated with strict confidentiality and in accordance with the Data Privacy Act.

We hope for your favorable consideration in this matter. Thank you very much, and God bless.

Respectfully yours,

**ARIAN JAY G. MAALA**

Group Leader

Noted by:

**TITO AMERIGO V. CUSTODIO JR., MIT**

Thesis Adviser

Recommending Approval:

**CHONA B. SABINAY, DIT**

Dean, STCS

Approved:

**GRETCHEN STEPHANIE M. ESPINA**

Municipal Mayor



## Appendix C

### LETTER OF REQUEST TO CONDUCT STUDY



Republic of the Philippines

**BILIRAN PROVINCE STATE UNIVERSITY**

ISO 9001:2015 CERTIFIED

**SCHOOL OF TECHNOLOGY AND COMPUTER STUDIES**

January 28, 2026

**GRETCHEN STEPHANIE M. ESPINA**

Municipal Mayor

Municipality of Naval

Brgy. P.I. Garcia, Naval, Biliran

Dear Ma'am:

The undersigned students respectfully request permission to conduct a study about the **Naval Public Market Price Watch (Bantay Presyo)** at the Agricultural Services Office of the Municipality of Naval.

We also respectfully call the attention of **Mr. Laurence Christian C. Mangco**, Municipal Agriculturist (MGDH-I), to provide the necessary data for the study. These include, but are not limited to, a soft copy of the list of monitored agricultural goods or commodities in the Naval Public Market, historical price data from 2019 to the present, and other pertinent materials for the development of the software. All information gathered will be treated with strict confidentiality in accordance with the Data Privacy Act of 2012.

We respectfully hope for your favorable consideration. Thank you very much.

Respectfully yours,

**ARIAN JAY G. MAALA**

Group Leader

Noted by:

**TITO AMERIGO V. CUSTODIO JR., MIT**

Thesis Adviser

Recommending Approval:

**CHONA B. SABINAY, DIT**

Dean, STCS

Approved:

**GRETCHEN STEPHANIE M. ESPINA**

Municipal Mayor

## Appendix D

### CONTRACT ADVISING



Republic of the Philippines

**BILIRAN PROVINCE STATE UNIVERSITY**

ISO 9001:2015 CERTIFIED

**SCHOOL OF TECHNOLOGY AND COMPUTER STUDIES**

### THESIS ADVISING CONTRACT

As this **THESIS ADVISER**, As the THESIS ADVISER, your primary role is to guide and support thesis or capstone project students by fulfilling the following specific responsibilities.

- Supervised and monitored the advisee's progress in developing the thesis proposal, conducting the study, and writing the final thesis report.
- Provide professional advice on the students' chosen thesis/capstone project topic and general support.
- Refer the advisee to other experts (e.g., content and research methods experts, data analysis, grammar editors) whom the advisee might consult.
- Bring to the advisee's attention any pertinent announcements regarding critical milestones and deadlines for T/D work, as well as funding opportunities.
- Facilitate, if necessary, the advisee's academic networking with local and international colleagues to gain access to and use of potential research materials, such as subscription journals and standardized research instruments.
- Regularly monitor the advisee's progress.
- Ascertain that all school and university requirements requiring committee actions are met on time by the student.

This contract serves as a legally binding document outlining your obligations and responsibilities to your thesis/capstone project advisee.

Signed:

**JINKY L. MACABAGO, PhD**

Dean, School of Technology and Computer Studies

I, **TITO AMERIGO V. CUSTODIO JR.**, from the **SCHOOL OF TECHNOLOGY AND COMPUTER STUDIES (STCS)**, pledge to fulfill my duties and responsibilities as **THESIS ADVISER** and to assist the students to the best of my ability in completing their research work.

Title: **SCHOLARSHIP RECORD MANAGEMENT WITH ELIGIBILITY RECOMMENDER FOR HIGHER EDUCATION INSTITUTION**

1. **ARIAN JAY G. MAALA**
2. **CARLVIN JAY G. SULLA**
3. **JUNIEL M. VILLABLANCA**
4. **SARAH JANE E. VILLANUEVA**

Signed:

**TITO AMERIGO V. CUSTODIO JR., MIT**

(Signature Over Printed Name of the Thesis Project Adviser)

## Appendix E

### LETTER OF REQUEST FOR DATA GATHERING



Republic of the Philippines

**BILIRAN PROVINCE STATE UNIVERSITY**

ISO 9001:2015 CERTIFIED

**SCHOOL OF TECHNOLOGY AND COMPUTER STUDIES**

August 5, 2026

**MARK RONNIE M. MANCO**

Director

Strategic Communications Office

Biliran Province State University

Dear Sir:

The undersigned are fourth-year Bachelor of Science in Computer Science students of Biliran Province State University, currently enrolled in the CS 4123 Thesis II Course.

We respectfully request permission to obtain soft copies of scholarship-related images and aerial photographs of the University from 2023–2026. These materials will be used for the user interface (UI) and content of our thesis project entitled **“Scholarship Record Management with Eligibility Recommender for Higher Education Institution.”**

Your approval and support will greatly contribute to the successful completion of our study.

Thank you, and God bless.

Respectfully yours,

**ARIAN JAY G. MAALA**

Group Leader

Noted by:

**TITO AMERIGO V. CUSTODIO JR., MIT**

Thesis Adviser

Recommending Approval:

**MARICEL S. SAULAN**

Scholarship In-Charge, SDSO

Approved:

**MARK RONNIE M. MANCO**

Director, Strategic Communications Office

## Appendix E

### LETTER OF REQUEST FOR DATA GATHERING



Republic of the Philippines

**BILIRAN PROVINCE STATE UNIVERSITY**

ISO 9001:2015 CERTIFIED

**SCHOOL OF TECHNOLOGY AND COMPUTER STUDIES**

August 5, 2026

**ALFRED JAMES O. ACEBEDO**

Chief Executive Editor

The Pillar Publication

Biliran Province State University

Dear Sir:

The undersigned are fourth-year Bachelor of Science in Computer Science students of Biliran Province State University, currently enrolled in the CS 4123 Thesis II Course.

We respectfully request permission to obtain soft copies of scholarship-related images and aerial photographs of the University from 2023–2026. These materials will be used for the user interface (UI) and content of our thesis project entitled **“Scholarship Record Management with Eligibility Recommender for Higher Education Institution.”**

Your approval and support will greatly contribute to the successful completion of our study.

Thank you, and God bless.

Respectfully yours,

**ARIAN JAY G. MAALA**

Group Leader

Noted by:

**TITO AMERIGO V. CUSTODIO JR., MIT**

Thesis Adviser

Recommending Approval:

**MARICEL S. SAULAN**

Scholarship In-Charge, SDSO

Approved:

**ALFRED JAMES O. ACEBEDO**

Chief Executive Editor, The Pillar Publication

## Appendix F

### DELIVERABLES AND MILESTONE

| TASKS                                                      | START     | END       | DURATION | 2026    |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
|------------------------------------------------------------|-----------|-----------|----------|---------|---|---|---|----------|---|---|---|-------|---|---|---|-------|---|---|---|-----|---|---|---|------|---|---|---|------|---|---|---|--------|---|---|---|-----------|---|---|---|---------|---|---|---|----------|---|---|---|----------|--|--|--|
|                                                            |           |           |          | JANUARY |   |   |   | FEBRUARY |   |   |   | MARCH |   |   |   | APRIL |   |   |   | MAY |   |   |   | JUNE |   |   |   | JULY |   |   |   | AUGUST |   |   |   | SEPTEMBER |   |   |   | OCTOBER |   |   |   | NOVEMBER |   |   |   | DECEMBER |  |  |  |
|                                                            |           |           |          | 1       | 2 | 3 | 4 | 1        | 2 | 3 | 4 | 1     | 2 | 3 | 4 | 1     | 2 | 3 | 4 | 1   | 2 | 3 | 4 | 1    | 2 | 3 | 4 | 1    | 2 | 3 | 4 | 1      | 2 | 3 | 4 | 1         | 2 | 3 | 4 | 1       | 2 | 3 | 4 | 1        | 2 | 3 | 4 |          |  |  |  |
| Manuscript Preparation                                     | Jan-1-26  | Jan-7-26  | 1 week   |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| Thesis Instructor Selection                                | Jan-7-26  | Jan-8-26  | 2 days   |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| Needs Assessment                                           | Jan-8-26  | Jan-22-26 | 2 weeks  |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| Requirements Gathering                                     | Jan-28-26 | TBA       |          |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| Requirements Analysis                                      | TBA       | TBA       |          |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| Title Hearing                                              | TBA       | TBA       |          |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| Working on Manuscript                                      | TBA       | TBA       |          |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| Data Gathering and Pre-Processing                          | TBA       | TBA       |          |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| Training and Building Model                                | TBA       | TBA       |          |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| Interview with Experts                                     | TBA       | TBA       |          |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| System Creation                                            | TBA       | TBA       |          |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| Data Gathering                                             | TBA       | TBA       |          |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| Continuation of Manuscript                                 | TBA       | TBA       |          |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| Pre-Oral Defense                                           | TBA       | TBA       |          |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| Working on Revisions (Results from Pre-Oral)               | TBA       | TBA       |          |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| Pre-Final Defense                                          | TBA       | TBA       |          |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| System Checking                                            | TBA       | TBA       |          |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| System Testing                                             | TBA       | TBA       |          |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| Final Defense                                              | TBA       | TBA       |          |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| Working on Manuscript (Recommendation from Panelist)       | TBA       | TBA       |          |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| Presentation of Revisions (Manuscript and System Checking) | TBA       | TBA       |          |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| Finalizing Manuscript & System                             | TBA       | TBA       |          |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| System Deployment                                          | TBA       | TBA       |          |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| System Evaluation                                          | TBA       | TBA       |          |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |
| Submission of Hardbound                                    | TBA       | TBA       |          |         |   |   |   |          |   |   |   |       |   |   |   |       |   |   |   |     |   |   |   |      |   |   |   |      |   |   |   |        |   |   |   |           |   |   |   |         |   |   |   |          |   |   |   |          |  |  |  |

## Appendix G

### BUDGET REQUIREMENTS

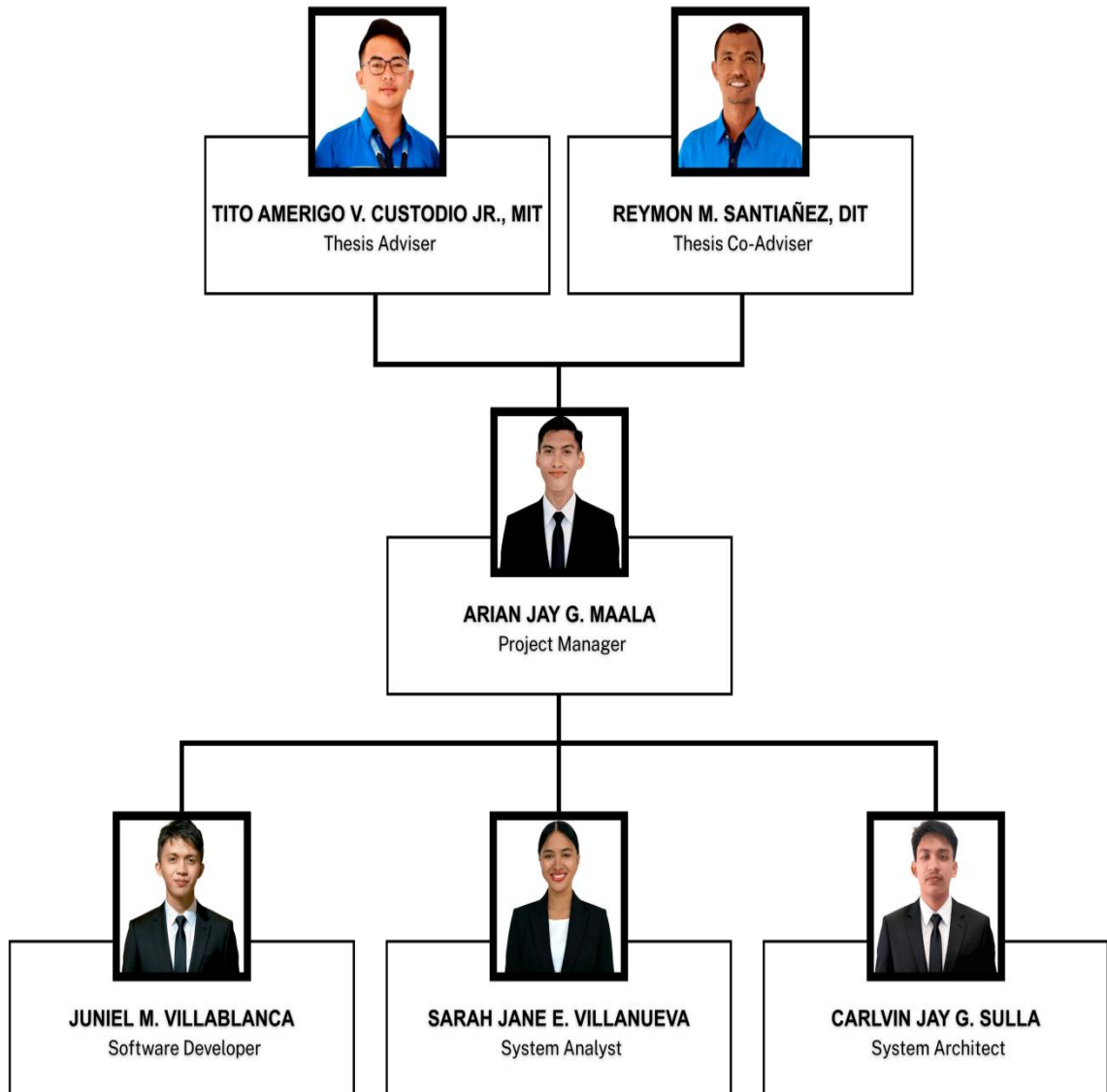
| Item                                                                    | Quantity | Unit Cost | Total Cost |
|-------------------------------------------------------------------------|----------|-----------|------------|
| <b>Supplies and Materials</b>                                           |          |           |            |
| Bond Paper                                                              | 2 rims   | 239       | 478        |
| Clear Book                                                              | 1 pc.    | 59        | 59         |
| Certificate Holder                                                      | 1 pc.    | 115       | 115        |
| Intermediate Pad                                                        | 1 pc.    | 60        | 60         |
| Sliding Folder                                                          | 10 pcs.  | 10        | 100        |
| <b>Meals and Snacks</b>                                                 |          |           |            |
| Snack for Group Members                                                 | 49       | 5         | 245        |
| Snacks and Meals for Title Hearing<br>(Per Group & Member Contribution) | 4        | 115       | 460        |
| Snacks and Meals for Title Hearing<br>(Personal Pledge)                 | 4        | 125       | 500        |
| Snacks and Meals for Pre-Oral &<br>Oral Defense                         | 4        | 92        | 368        |
| Snacks and Meals for Pre-Final<br>Defense                               |          |           |            |
| Snacks and Meals for Final Defense                                      |          |           |            |
| <b>Travel Fee</b>                                                       |          |           |            |
| Tricycle                                                                | 31       | 10        | 310        |
| <b>Thesis Group Uniform</b>                                             | 4        | 600       | 2400       |
| <b>Online Checker</b>                                                   |          |           |            |

|                      |  |                     |  |
|----------------------|--|---------------------|--|
| Turnitin             |  |                     |  |
| Grammarly            |  |                     |  |
| <b>OVERALL TOTAL</b> |  | <b>PHP 5,095.00</b> |  |

## Appendix H

### PROJECT TEAM ORGANIZATION AND WORK TASK

#### THESIS PROJECT ORGANIZATIONAL STRUCTURE





## PROJECT TEAM WORK TASK

**TITO AMERIGO V. CUSTODIO JR., MIT** (*Thesis Adviser*). As the Thesis Adviser, his role is to guide and support the student project team throughout the thesis process. He provides academic guidance on research design, methodology, data analysis, and overall project planning. The adviser helps that the project meets academic standards, offers constructive feedback, and mentors students in navigating challenges and developing the necessary skills to complete a successful thesis.

**MARICEL S. SAULAN** (*Thesis Co-Adviser*). The Thesis Co-Adviser assists the Thesis Adviser in providing academic and technical support. This role supports students with specialized expertise as needed, participates in meetings with the team, and contributes to feedback on research progress and written work. The co-adviser works collaboratively with the primary adviser to help students achieve the project's research objectives.

**ARIAN JAY G. MAALA** (*Project Manager*). The Project Manager is responsible for coordinating and overseeing the execution of the project. This role includes defining objectives, organizing team tasks, managing schedules and deliverables, and ensures that project goals are achieved within the set timeline and scope. The project manager also facilitates communication among team members and resolves issues that may affect project progress.

**JUNIEL M. VILLABLANCA** (*Software Developer*). The Software Developer designs and writes the code needed to build the software solution defined in the project. This role involves translating system requirements into functional software components, testing features to ensure the system work

as expected, and working closely with other technical team members to implement and refine software functionality.

**SARAH JANE E. VILLANUEVA** (*System Analyst*). The Systems Analyst analyzes user needs and system requirements to design solutions that satisfy those needs. This includes defining functional requirements, identifying opportunities for improvements in processes or software designs, and ensures that the technical implementation aligns with project goals. Systems analysts also act as liaisons between users, developers, and project managers.

**CARLVIN JAY G. SULLA** (*System Architect*). The System Architect defines the high-level structure of the system, breaking it into components and specifying how those components work together to meet project requirements. This role involves making key technical decisions about software structure, choosing appropriate technologies, and ensure the overall design supports performance, scalability, security, and maintainability.

## **Appendix I**

### **RESOURCES**

## APPENDIX J

### SURVEY INSTRUMENT



Republic of the Philippines  
BILIRAN PROVINCE STATE UNIVERSITY  
ISO 9001:2015 CERTIFIED  
SCHOOL OF TECHNOLOGY AND COMPUTER STUDIES



Dear Respondent,

The undersigned are currently in the process of evaluating the functionality of the **“SCHOLARSHIP RECORD MANAGEMENT WITH ELIGIBILITY RECOMMENDER FOR HIGHER EDUCATION INSTITUTION SYSTEM”** as part of the requirement in partial fulfillment of the degree of Bachelor of Science in Computer Science (BSCS).

We earnestly ask for your response to the questions provided below. Rest assured that the data collected through this tool will be used for academic purposes only and will be treated with utmost confidentiality.

Respectfully yours,

Name (optional): \_\_\_\_\_

Position: \_\_\_\_\_

Directions. Please review the relevant technical components of the SCHOLARSHIP RECORD MANAGEMENT WITH ELIGIBILITY RECOMMENDER FOR HIGHER EDUCATION INSTITUTION SYSTEM and rate each statement on the level of compliance with software quality standards principles in terms of: **Functional Suitability, Performance Efficiency, Compatibility, Interaction Capability, Reliability, Security, Maintainability, Flexibility, and Safety.**

Please assess the system by checking the number that corresponds to your rating as specified below:

**5 – Very High Extent**

**4 – High Extent**

**3 – Moderate Extent**

**2 – Low Extent**

**1 – Very Low Extent**

**Objective 4. Evaluate the developed system using white box testing and the ISO 25010**

| Clarity       |            | Relevance | Comments/  |
|---------------|------------|-----------|------------|
| 1 – Not Clear | Indicators | 1 – Not   | Suggestion |

|                                                                                                                        |   |   |   |                                                                                                                                                                         |                                                                                                                                                     |          |   |   |  |
|------------------------------------------------------------------------------------------------------------------------|---|---|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------|---|---|--|
| <p><b>2 – Item need some revision</b></p> <p><b>3 – Clear but need minor revision</b></p> <p><b>4 – Very clear</b></p> |   |   |   |                                                                                                                                                                         | <p><i>relevant</i></p> <p><b>2 – Item need some revision</b></p> <p><b>3 – Relevant but need minor revision</b></p> <p><b>4 – Very Relevant</b></p> | <b>s</b> |   |   |  |
| 1                                                                                                                      | 2 | 3 | 4 | <b>ISO 25010 Software Quality Standards</b>                                                                                                                             | 1                                                                                                                                                   | 2        | 3 | 4 |  |
|                                                                                                                        |   |   |   | <b>A. Functional Suitability</b>                                                                                                                                        |                                                                                                                                                     |          |   |   |  |
|                                                                                                                        |   |   |   | <p><b>1. Functional Completeness.</b><br/>Capability of a product to provide a set of functions that covers all the specified tasks and intended users' objectives.</p> |                                                                                                                                                     |          |   |   |  |
|                                                                                                                        |   |   |   | <p><b>2. Functional Correctness.</b><br/>Capability of a product to provide accurate results when used by intended users</p>                                            |                                                                                                                                                     |          |   |   |  |
|                                                                                                                        |   |   |   | <p><b>3. Functional Appropriateness.</b><br/>Capability of a product to provide functions that facilitate the accomplishment of specified tasks and objectives.</p>     |                                                                                                                                                     |          |   |   |  |
|                                                                                                                        |   |   |   | <b>B. Performance efficiency</b>                                                                                                                                        |                                                                                                                                                     |          |   |   |  |
|                                                                                                                        |   |   |   | <b>1. Time Behaviour.</b>                                                                                                                                               |                                                                                                                                                     |          |   |   |  |

|  |  |  |  |                                                                                                                                                                                                                                                                                                                        |  |  |  |  |  |
|--|--|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|  |  |  |  | Capability of a product to perform its specified function under specified conditions so that the response time and throughput rates meet the requirements.                                                                                                                                                             |  |  |  |  |  |
|  |  |  |  | <p>2. <b>Resource Utilization.</b></p> <p>Capability of a product to use no more than the specified amount of resources to perform its function under specified conditions.</p>                                                                                                                                        |  |  |  |  |  |
|  |  |  |  | <p>3. <b>Capacity.</b> Capability of a product to meet requirements for the maximum limits of a product parameter (<i>Parameters can include the number of items that can be stored, the number of concurrent users, the communication bandwidth, the throughput of transactions, and the size of a database</i>).</p> |  |  |  |  |  |
|  |  |  |  | <b>C. Compatibility</b>                                                                                                                                                                                                                                                                                                |  |  |  |  |  |
|  |  |  |  | <p>1. <b>Co-existence.</b></p> <p>Capability of a product to perform its required functions efficiently while sharing a common environment and resources with other products, without detrimental impact</p>                                                                                                           |  |  |  |  |  |

|  |  |  |  |                                                                                                                                                              |  |  |  |  |  |
|--|--|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|  |  |  |  | on any other product.                                                                                                                                        |  |  |  |  |  |
|  |  |  |  | <b>2. Interoperability.</b><br>Capability of a product to exchange information with other products and mutually use the information that has been exchanged. |  |  |  |  |  |
|  |  |  |  | <b>D. Interaction capability</b>                                                                                                                             |  |  |  |  |  |
|  |  |  |  | <b>1. Appropriateness Recognizability.</b><br>Capability of a product to be recognized by users as appropriate for their needs.                              |  |  |  |  |  |
|  |  |  |  | <b>2. Learnability.</b><br>Capability of a product to have specified users learn to use specified product functions within a specified amount of time.       |  |  |  |  |  |
|  |  |  |  | <b>3. Operability.</b><br>Capability of a product to have functions and attributes that make it easy to operate and control.                                 |  |  |  |  |  |
|  |  |  |  | <b>4. User Error Protection.</b><br>Capability of a product to prevent operation errors.                                                                     |  |  |  |  |  |
|  |  |  |  | <b>5. User Engagement.</b><br>Capability of a product to present functions and information in an                                                             |  |  |  |  |  |

|  |  |  |  |                                                                                                                                                                                                                                                             |  |  |  |  |  |
|--|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|  |  |  |  | inviting and motivating manner encouraging continued interaction.                                                                                                                                                                                           |  |  |  |  |  |
|  |  |  |  | <b>6. User Assistance.</b><br>Capability of a product to be used by people with the widest range of characteristics and capabilities to achieve specified goals in a specified context of use.                                                              |  |  |  |  |  |
|  |  |  |  | <b>7. Inclusivity.</b><br>Capability of a product to be utilised by people of various backgrounds.                                                                                                                                                          |  |  |  |  |  |
|  |  |  |  | <b>8. Self-descriptiveness.</b><br>Capability of a product to present appropriate information, where needed by the user, to make its capabilities and use immediately obvious to the user without excessive interactions with a product or other resources. |  |  |  |  |  |
|  |  |  |  | <b>E. Reliability</b>                                                                                                                                                                                                                                       |  |  |  |  |  |
|  |  |  |  | <b>1. Faultlessness.</b><br>Capability of a product to perform specified functions without fault under normal operation                                                                                                                                     |  |  |  |  |  |



|  |  |  |  |                                                                                                                                                                                                     |  |  |  |  |  |
|--|--|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|  |  |  |  | <b>2. Availability.</b><br>Capability of a product to be operational and accessible when required for use.                                                                                          |  |  |  |  |  |
|  |  |  |  | <b>3. Fault Tolerance.</b><br>Capability of a product to operate as intended despite the presence of hardware or software faults.                                                                   |  |  |  |  |  |
|  |  |  |  | <b>4. Recoverability.</b><br>Capability of a product in the event of an interruption or a failure to recover the data directly affected and re-establish the desired state of the system.           |  |  |  |  |  |
|  |  |  |  | <b>F. Security</b>                                                                                                                                                                                  |  |  |  |  |  |
|  |  |  |  | <b>1. Confidentiality.</b><br>Capability of a product to ensure that data are accessible only to those authorized to have access.                                                                   |  |  |  |  |  |
|  |  |  |  | <b>2. Integrity.</b> Capability of a product to ensure that the state of its system and data are protected from unauthorized modification or deletion either by malicious action or computer error. |  |  |  |  |  |
|  |  |  |  | <b>3. Non-repudiation.</b><br>Capability of a product to prove that                                                                                                                                 |  |  |  |  |  |

|  |  |  |  |                                                                                                                              |  |  |  |  |  |
|--|--|--|--|------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|  |  |  |  | actions or events have taken place, so that the events or actions cannot be repudiated later.                                |  |  |  |  |  |
|  |  |  |  | 4. <b>Accountability.</b><br>Capability of a product to enable actions of an entity to be traced uniquely to the entity.     |  |  |  |  |  |
|  |  |  |  | 5. <b>Authenticity.</b><br>Capability of a product to prove that the identity of a subject or resource is the one claimed.   |  |  |  |  |  |
|  |  |  |  | 6. <b>Resistance.</b><br>Capability of a product to sustain operations while under attack from a malicious actor.            |  |  |  |  |  |
|  |  |  |  | <b>G. Maintainability</b>                                                                                                    |  |  |  |  |  |
|  |  |  |  | 1. <b>Modularity.</b><br>Capability of a product to limit changes to one component from affecting other components.          |  |  |  |  |  |
|  |  |  |  | 2. <b>Reusability.</b><br>Capability of a product to be used as assets in more than one system, or in building other assets. |  |  |  |  |  |
|  |  |  |  | 3. <b>Analysability.</b><br>Capability of a product to be effectively and                                                    |  |  |  |  |  |

|  |  |  |  |                                                                                                                                                                                             |  |  |  |  |  |
|--|--|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|  |  |  |  | efficiently assessed regarding the impact of an intended change to one or more of its parts, to diagnose it for deficiencies or causes of failures, or to identify parts to be modified.    |  |  |  |  |  |
|  |  |  |  | <b>4. Testability.</b><br>Capability of a product to enable an objective and feasible test to be designed and performed to determine whether a requirement is met.                          |  |  |  |  |  |
|  |  |  |  | <b>H. Flexibility</b>                                                                                                                                                                       |  |  |  |  |  |
|  |  |  |  | <b>1. Adaptability.</b><br>Capability of a product to be effectively and efficiently adapted for or transferred to different hardware, software or other operational or usage environments. |  |  |  |  |  |
|  |  |  |  | <b>2. Scalability.</b><br>Capability of a product to handle growing or shrinking workloads or to adapt its capacity to handle variability.                                                  |  |  |  |  |  |
|  |  |  |  | <b>3. Installability.</b><br>Capability of a product to be effectively and efficiently installed successfully and/or                                                                        |  |  |  |  |  |

|  |  |  |  |                                                                                                                                                                            |  |  |  |  |  |
|--|--|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|  |  |  |  | uninstalled in a specified environment.                                                                                                                                    |  |  |  |  |  |
|  |  |  |  | <b>4. Replaceability.</b><br>Capability of a product to replace another specified product for the same purpose in the same environment.                                    |  |  |  |  |  |
|  |  |  |  | <b>I. Safety</b>                                                                                                                                                           |  |  |  |  |  |
|  |  |  |  | <b>1. Operational Constraint.</b><br>Capability of a product to constrain its operation to within safe parameters or states when encountering operational hazard.          |  |  |  |  |  |
|  |  |  |  | <b>2. Risk Identification.</b><br>Capability of a product to identify a course of events or operations that can expose life, property or environment to unacceptable risk. |  |  |  |  |  |
|  |  |  |  | <b>3. Fail Safe.</b> Capability of a product to automatically place itself in a safe operating mode, or to revert to a safe condition in the event of a failure.           |  |  |  |  |  |
|  |  |  |  | <b>4. Hazard Warning.</b><br>Capability of a product to provide warnings of unacceptable risks to operations or internal controls so that they                             |  |  |  |  |  |

|  |  |  |  |                                                                                                                                     |  |  |  |  |  |
|--|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|  |  |  |  | can react in sufficient time to sustain safe operations.                                                                            |  |  |  |  |  |
|  |  |  |  | 5. <b>Safe Integration.</b><br>Capability of a product to maintain safety during and after integration with one or more components. |  |  |  |  |  |

What would you suggest to help improve the Scholarship Record Management with Eligibility Recommender for Higher Education Institution System?

---



---



---

I hereby affix my signature as a sign of my evaluation in using the system being tested.

Confirmed:

---

Name and Signature of the Respondent

## **Appendix K**

### **CONTRIBUTORS**

These are the individuals who played a significant role in the success of this study by providing their expertise, guidance, and support throughout the research and development process.

#### **Resource Persons**

**DR. CHONA B. SABINAY** - Dean, School of Technology and Computer Studies

**MR. TITO AMERIGO V. CUSTODIO JR.** - MIT, Thesis Adviser

**DR. REYMON M. SANTIAÑEZ** - Thesis Instructor, Co-Adviser

## **Appendix L**

### **PROGRAM LISTING**

## **Appendix M**

### **TECHNICAL REFERENCES**



## **Appendix N**

### **USER AND OPERATIONAL MANUAL**

## **Appendix O**

### **SOFTWARE REQUIREMENT AND SPECIFICATION**

## Appendix P

### PROJECT DEVELOPMENT AND EVALUATION DOCUMENTATION

The documentation for a thesis project on Scholarship Record Management with Eligibility Recommender for Higher Education Institution may include the following:

1. Thesis Adviser Selection - The process of selecting and assigning a qualified thesis adviser who provides guidance, supervision, and technical expertise throughout the planning, development, implementation, and evaluation of the research project.



2. Permission to Conduct a Needs Assessment - An approved request to gather information through surveys, interviews, and other data-gathering activities necessary to identify the current challenges and requirements of the scholarship management process.



3. Project Proposal - A comprehensive proposal that outlines the problem statement, objectives, scope, methodology, timeline, and expected outcomes of the project.



4. Literature Review - A thorough review of relevant literature and studies on prison management systems, including their features, benefits, and limitations.



5. Title Hearing - A formal presentation of the proposed research title, objectives, significance, and scope before the panel for evaluation, recommendations, and approval.





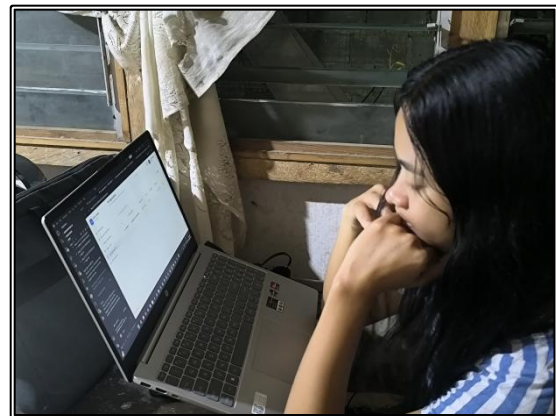
6. Data gathering and Pre-processing - The process of collecting relevant data through surveys, interviews, institutional records, and existing scholarship guidelines, followed by cleaning, organizing, validating, and preparing the data for system analysis and development.



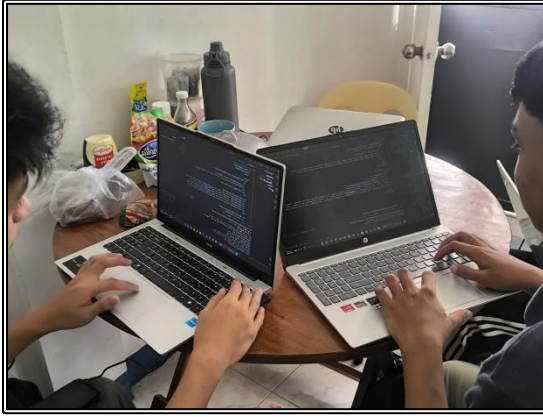
7. **Pre-oral & Oral Defense** - Formal presentations of the research proposal before the evaluation panel to validate the study's objectives, methodology, feasibility, and overall research design, while incorporating recommendations for improvement.



8. System Requirements Specification (SRS) - A detailed documents that outlines the functional and non-functional requirements of the system, including user requirements, and performance metrics.



9. Application Development - The process of designing, coding, integrating, and implementing the Scholarship Record Management with Eligibility Recommender System based on the approved system requirements and design specifications.

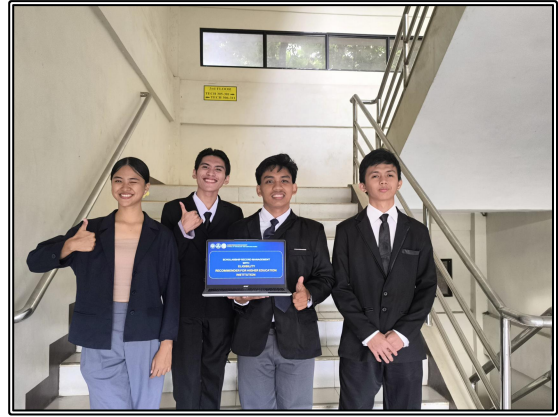


10. Design Document - A document that describes the overall system architecture, database design, entity-relationship diagrams, user interface layouts, system modules, process workflows, and design standards followed during development.



11. Pre-final Defense - A presentation of the completed system prototype and documentation before the panel to evaluate the system's functionality, documentation quality, and readiness for final implementation, while addressing recommended revisions.



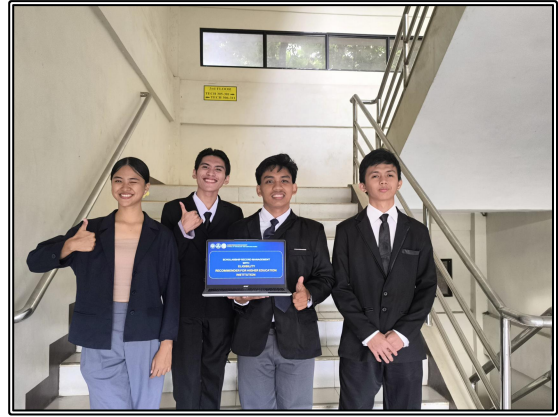


12. **Implementation Document** - A document that provides a step-by-step guide on how the system was implemented, including the programming languages, frameworks, and libraries used.



13. **Testing & Evaluation** - A report that outlines the testing procedures, test cases, and evaluation metrics used to assess the functionality, reliability, and usability of the system.

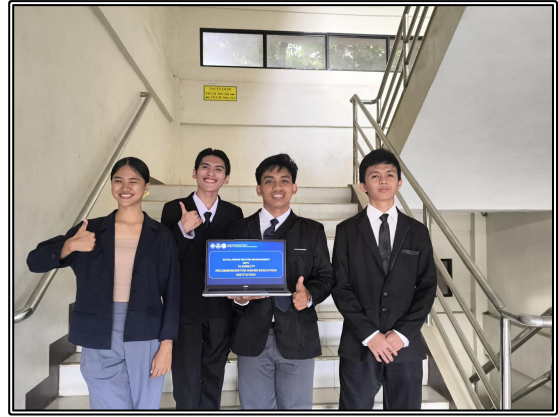




14. **Final Defense** - The final presentation and demonstration of the completed research project before the evaluation panel, showcasing the developed system, research findings, recommendations, and compliance with all academic requirements.



15. **User Manual** - A comprehensive guide that provides users with step-by-step instructions for operating the system, including logging in, managing scholarship records, evaluating student eligibility, generating reports, troubleshooting common issues, and obtaining technical support.



16. **Project Report** - The final documentation of the research project that summarizes the study's background, objectives, methodology, system development process, implementation, testing results, conclusions, recommendations, and future enhancements.



## **CURRICULUM VITAE**

**ARIAN JAY G. MAALA**

Looc, Cabucgayan, Biliran

arianmaala19@gmail.com

09544856075

**OBJECTIVE**

To apply my knowledge and skills in computer science to develop innovative, efficient, and user-centered digital solutions. I strive to produce high-quality and impactful outputs that enhance productivity, empower institutions, and create meaningful improvements in the lives of people and communities across diverse backgrounds.

**PERSONAL INFORMATION**

|                |   |                                       |
|----------------|---|---------------------------------------|
| Sex            | : | Male                                  |
| Date of Birth  | : | October 2, 2005                       |
| Place of Birth | : | BPH - Naval, Biliran                  |
| Nationality    | : | Filipino                              |
| Civil Status   | : | Single                                |
| Language       | : | English/ Tagalog/ Waray-Waray/ Bisaya |

**EDUCATIONAL ATTAINMENT**

|           |   |                                                                                                                                                                 |
|-----------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TERTIARY  | : | Biliran Province State University<br>School of Technology and Computer Studies<br>Bachelor of Science in Computer Science<br>P.I. Garcia Street, Naval, Biliran |
| ACADEMICS | : | Cabucgayan National High School (CNHS)<br>Accountancy, Business, and Management (ABM)<br>Esperanza, Cabucgayan, Biliran<br>S.Y 2022-2023                        |
| SECONDARY | : | Cabucgayan National High School (CNHS)                                                                                                                          |

Esperanza, Cabucgayan, Biliran  
S.Y 2020-2021

ELEMENTARY : Looc Elementary School  
Looc, Cabucgayan, Biliran  
S.Y 2016-2017

### **QUALIFICATION SKILLS**

- Computer Literate
- Researching and Mining Data
- Proficient in editing using Canva and Capcut
- Proficient in using Microsoft Office (Word, Powerpoint, Publisher)

### **CHARACTER REFERENCES**

#### **VERONICA SHANE O. GENOGUIN**

Peer

Barangay Casiawan  
Cabucgayan, Biliran

#### **VICTORINO M. REFUERZO**

Teacher

Barangay Esperanza  
Cabucgayan, Biliran

**CARLVIN JAY G. SULLA**

Looc, Cabucgayan, Biliran

sullacarlvinjau@gmail.com

09498786027

**OBJECTIVE**

To continuously enhance my knowledge in Computer Science and be able to apply my skills to contribute more effectively..

**PERSONAL INFORMATION**

|                |   |                                   |
|----------------|---|-----------------------------------|
| Sex            | : | Male                              |
| Date of Birth  | : | August 11, 2004                   |
| Place of Birth | : | Looc, Cabucgayan, Biliran         |
| Nationality    | : | Filipino                          |
| Civil Status   | : | Single                            |
| Language       | : | English/ Tagalog/ Cebuano/ Bisaya |

**EDUCATIONAL ATTAINMENT**

|           |   |                                                                                                                                                                      |
|-----------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TERTIARY  | : | Biliran Province State University<br>School of Technology and Computer Studies<br>Bachelor of Science in Computer Science<br>P.I. Garcia Street, Naval, Biliran      |
| ACADEMICS | : | Cabucgayan National School of Arts and Sciences (CNSAT)<br>Science, Technology, Engineering and Mathematics (STEM)<br>Libertad, Cabucgayan, Biliran<br>S.Y 2022-2023 |
| SECONDARY | : | Cabucgayan National High School (CNHS)<br>Esperanza, Cabucgayan, Biliran                                                                                             |

S.Y 2020-2021

ELEMENTARY : Looc Elementary School  
Looc, Cabucgayan, Biliran  
S.Y 2016-2017

### QUALIFICATION SKILLS

- Computer Literate
- Proficient in using Canva editing tool

### CHARACTER REFERENCES

#### **RYAN C. GARCIA**

Teacher

Barangay Bliss

Naval, Biliran

#### **ARIAN JAY G. MAALA**

Peer

Barangay Looc

Cabucgayan, Biliran

**JUNIEL M. VILLABLANCA**

Mapuyo, Kawayan, Biliran

junielvillablanca2005@gmail.com

09539864154

**OBJECTIVE**

To continuously enhance my knowledge in computer science and be able to apply my skills to contribute more effectively.

**PERSONAL INFORMATION**

|                |   |                                   |
|----------------|---|-----------------------------------|
| Sex            | : | Male                              |
| Date of Birth  | : | May 13, 2005                      |
| Place of Birth | : | Kawayan, Biliran                  |
| Nationality    | : | Filipino                          |
| Civil Status   | : | Single                            |
| Language       | : | English/ Tagalog/ Cebuano/ Bisaya |

**EDUCATIONAL ATTAINMENT**

|           |   |                                                                                                                                                                 |
|-----------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TERTIARY  | : | Biliran Province State University<br>School of Technology and Computer Studies<br>Bachelor of Science in Computer Science<br>P.I. Garcia Street, Naval, Biliran |
| ACADEMICS | : | Kawayan National High School (KNHS)<br>General Academic Strand (GAS)<br>Poblacion, Kawayan, Biliran<br>S.Y 2022-2023                                            |
| SECONDARY | : | Kawayan National High School (KNHS)<br>Poblacion, Kawayan, Biliran<br>S.Y 2020-2021                                                                             |



ELEMENTARY : Mapuyo Elementary School  
Mapuyo, Kawayan, Biliran  
S.Y 2016-2017

### **QUALIFICATION SKILLS**

- Computer Literate
- Proficient in using Microsoft Office (Word, Powerpoint, Publisher)

### **CHARACTER REFERENCES**

#### **GRACE I. ESTREMOS**

Teacher

Barangay Mapuyo

Kawayan, Biliran

#### **APRIL C. RAFAEL**

Teacher

Barangay Mapuyo

Kawayan, Biliran

**SARAH JANE E. VILLANUEVA**

Masagaosao, Kawayan, Biliran

villanuevasarahjane59@gmail.com

09653764660

**OBJECTIVE**

One of my main goal is not just to develop and create software application but to gain experiences and explore things beyond the classroom and learn from that.

**PERSONAL INFORMATION**

|                |   |                            |
|----------------|---|----------------------------|
| Sex            | : | Female                     |
| Date of Birth  | : | January 9, 2005            |
| Place of Birth | : | Naval, Biliran             |
| Nationality    | : | Filipino                   |
| Civil Status   | : | Single                     |
| Language       | : | Cebuano/ Filipino/ English |

**EDUCATIONAL ATTAINMENT**

|           |   |                                                                                                                                                                 |
|-----------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TERTIARY  | : | Biliran Province State University<br>School of Technology and Computer Studies<br>Bachelor of Science in Computer Science<br>P.I. Garcia Street, Naval, Biliran |
| ACADEMICS | : | Kawayan National High School (KNHS)<br>General Academic Strand (GAS)<br>Poblacion, Kawayan, Biliran<br>S.Y 2022-2023                                            |
| SECONDARY | : | Kawayan National High School (KNHS)<br>Poblacion, Kawayan, Biliran<br>S.Y 2020-2021                                                                             |

ELEMENTARY : Kawayan Elementary School  
Masagaosao, Naval, Biliran  
S.Y 2016-2017

### **QUALIFICATION SKILLS**

- Computer Literate
- Proficient in using Microsoft Office (Word, Powerpoint, Publisher)

### **CHARACTER REFERENCES**

**RAYMUNDO M. EBAJO Jr.**

Barangay Captain

Barangay Masagaosao

Kawayan, Biliran

**ARIAN JAY G. MAALA**

Peer

Barangay Looc

Cabucgayan, Biliran